

Pearson Edexcel Level 3 GCE

Monday 23 June 2025

Afternoon (Time: 1 hour 30 minutes)

Paper
reference

9FM0/4D

Further Mathematics

Advanced

PAPER 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
- – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
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1. Ravi can choose one of three options, A, B or C, when playing a game. The profit, in pounds, associated with each outcome, and the corresponding probabilities, are shown in the decision tree in Figure 1.

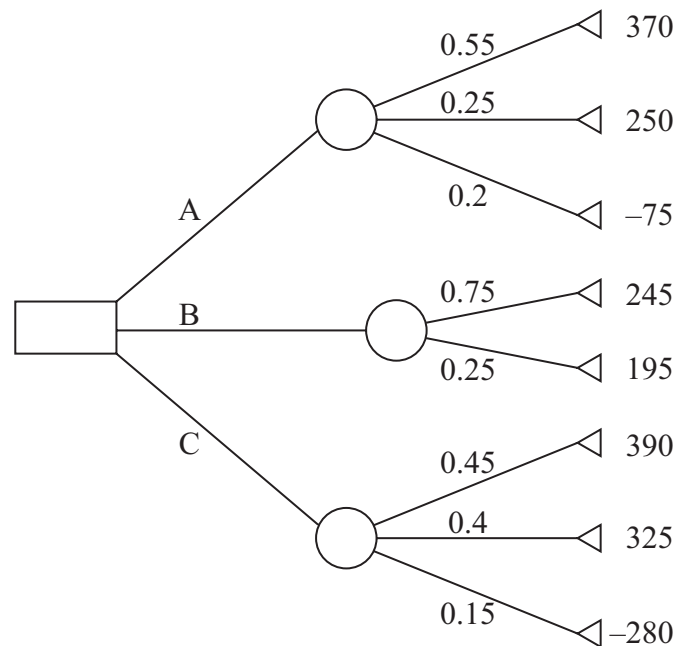


Figure 1

Calculate the optimal EMV to determine Ravi's best course of action. You must make your working clear.

(Total for Question 1 is 3 marks)

2. Five workers, A, B, C, D and E, are available to complete four tasks, P, Q, R and S.

Each worker can be assigned to at most one task, and each task must be done by at most one worker.

Worker B cannot be assigned to task R.

The time, in minutes, that each worker takes to complete each task is shown in the table below.

	P	Q	R	S
A	25	32	43	28
B	41	37	–	38
C	43	35	37	39
D	40	33	37	41
E	37	38	43	40

The Hungarian algorithm is to be used to find an allocation that minimises the total time to complete all four tasks.

- (a) Explain how the table should be modified so that the Hungarian algorithm can be applied. (2)
- (b) Use the Hungarian algorithm to obtain an allocation that minimises the total time. (6)
- (c) Calculate the least total time to complete all four tasks. (1)

(Total for Question 2 is 9 marks)

3. The table below shows the cost, in pounds, of transporting one unit of stock from each of four supply points, A, B, C and D, to three demand points, P, Q and R. It also shows the stock held at each supply point and the number of units required at each demand point.

A minimum cost solution is required.

	P	Q	R	Supply
A	27	23	25	35
B	29	30	28	41
C	29	33	26	29
D	32	34	36	45
Demand	57	31	62	

- (a) Use the north-west corner method to obtain an initial solution to this transportation problem. (1)
- (b) Perform one iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by showing the route and stating the
- shadow costs
 - improvement indices
 - entering cell and exiting cell
- (4)
- (c) Formulate the transportation problem as a linear programming problem. You must define your decision variables and make the objective function and constraints clear. (5)

(Total for Question 3 is 10 marks)

4.

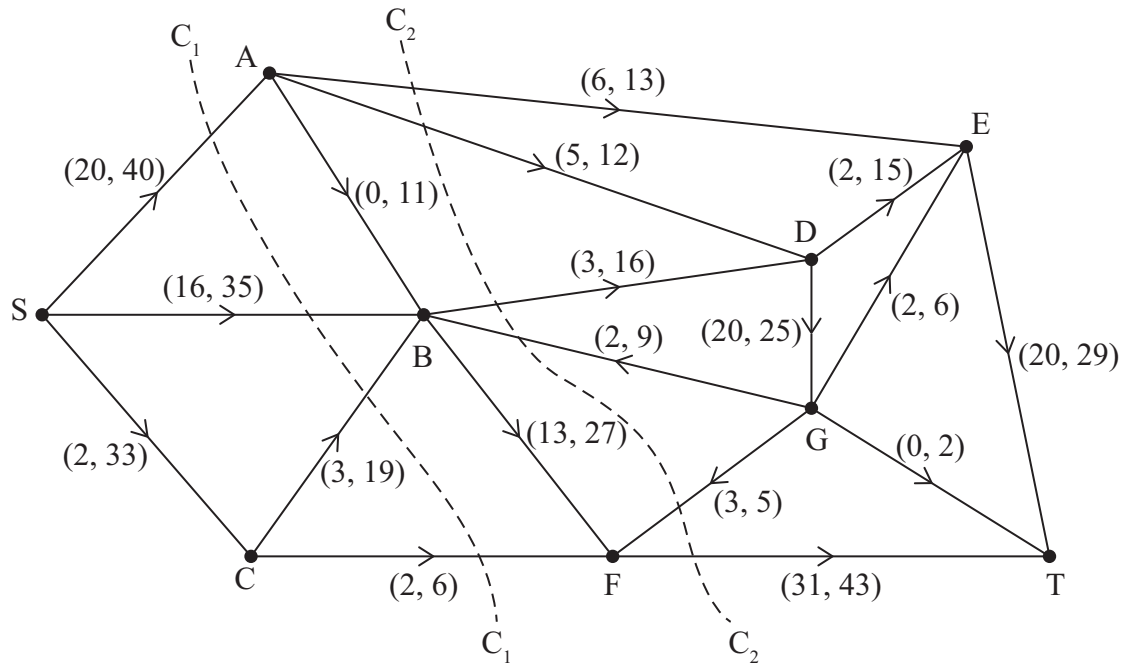


Figure 2

Figure 2 shows a capacitated, directed network. The network represents a system of pipes through which fluid can flow.

The weights on the arcs show the lower and upper capacities for the corresponding pipes, in litres per second.

(a) Calculate the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(b) Using only the capacities of cuts C_1 and C_2 , state what can be deduced about the maximum flow through the system.

(1)

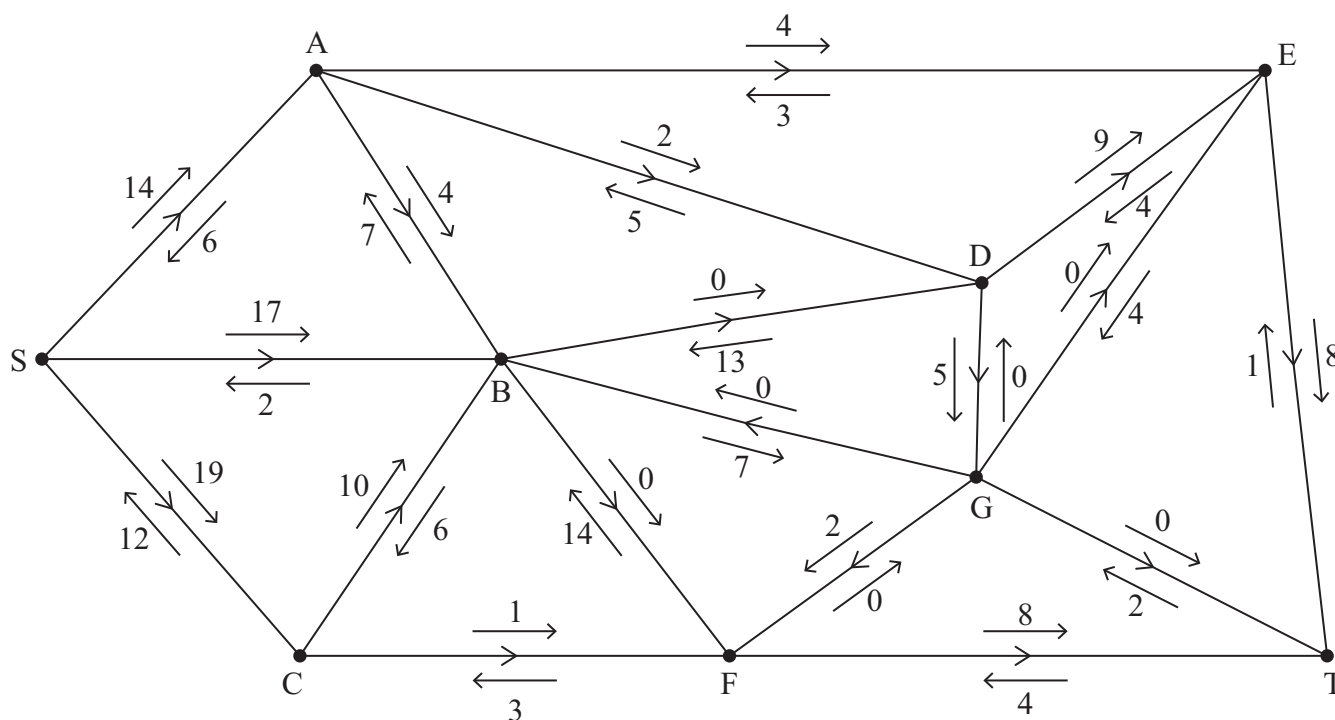


Figure 3

A scientist monitors the system of pipes in which a fluid now flows from the source, S, to the sink, T.

The scientist initialises the labelling procedure for this system. The excess and potential backflows for this are shown on the arrows either side of each arc in Figure 3.

- (c) State the value of the initial flow. (1)
- (d) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow. (3)
- (e) Use your answer to part (d) to find a maximum flow pattern for this system of pipes and draw it on Diagram 1 in the answer book. (1)
- (f) Prove that the answer to part (e) is optimal. (3)

(Total for Question 4 is 11 marks)

5. Anvi makes boats during the winter months. She can make up to five boats each month.

If she builds more than three boats in any one month, she must hire an assistant at a cost of £400 for that month.

In any month in which boats are made, the overhead costs are £60 for each boat made that month.

A maximum of three boats can be held in storage at the end of each month, at a cost of £70 per boat per month.

Boats must be delivered at the end of each month.

The order schedule for boats is

Month	November	December	January	February	March
Number ordered	2	5	6	4	3

There are no boats in storage at the beginning of November.

Anvi plans to have no boats left in storage after the end of the March delivery.

- (a) Use dynamic programming to determine the production schedule that minimises the costs given above. Complete the working in the table provided in the answer book and state the minimum cost.

(12)

The assistant asks to work in November, and Anvi agrees.

- (b) Given that more than three boats are now made in November, state the minimum additional cost of producing all the required boats to the schedule shown above.

(1)

(Total for Question 5 is 13 marks)

6.

		Player B		
		Option P	Option Q	Option R
Player A	Option X	4	-2	-5
	Option Y	-1	1	3

A two person zero-sum game is represented by the pay-off matrix for player A shown above.

- (a) Verify that there is no stable solution to this game. (2)

Let player A play option X with probability p

- (b) Use a graphical method to find the optimal value of p and hence find the best strategy for player A in this game. (6)

A third option, Z, is added to player A's options. Option Z has the pay-offs shown in the matrix below.

		Player B		
		Option P	Option Q	Option R
Player A	Option X	4	-2	-5
	Option Y	-1	1	3
	Option Z	4	-1	1

Player A now intends to make a random choice between options X, Y and Z, choosing option X with probability x , option Y with probability y and option Z with probability z

Player A decides to use the Simplex algorithm to find the optimal values of x , y and z

- (c) Determine an initial Simplex tableau to solve this 3×3 game, making your variables clear. (4)

In the optimal Simplex tableau for this 3×3 game, $x = 0$ and $y = \frac{5}{7}$

- (d) By considering the values of the two games, determine how much better the 3×3 game (with option Z) is for player A compared with the 2×3 game (without option Z).
You must make your reasoning clear. (3)

(Total for Question 6 is 15 marks)

7. A sequence $\{u_n\}$, where $n \geq 0$, satisfies the first order recurrence relation (*)

$$u_{n+1} = \alpha u_n - \beta(k^n) \quad (*)$$

where α, β, k are non-zero constants and $\alpha \neq k$

(a) Determine, in terms of α, β, k and n , the general solution of this recurrence relation. (4)

Given that $u_0 = L$ and $u_T = 0$, where T is a constant,

(b) show that a particular solution of the recurrence relation is

$$u_n = \frac{Lk^n \left(1 - \left(\frac{k}{\alpha} \right)^{T-n} \right)}{1 - \left(\frac{k}{\alpha} \right)^T} \quad (5)$$

Taylor borrows a sum of money, L , to buy a house.

Taylor arranges a loan which enables them to increase their repayment as their income grows.

The terms of the loan require Taylor's repayments to be made once a year.

On each anniversary of the loan being taken out, 6% interest is charged on the outstanding balance and Taylor makes an annual repayment of the total loan.

Taylor's total annual repayment of the loan will increase by 4% each year and the whole debt will be cleared in exactly 40 years.

The recurrence relation (*) can be used to model this situation.

(c) (i) State the value of α
(ii) State the value of k
(iii) State the relevance of β in (*) (3)

(d) Show that after 30 years, according to the model, the outstanding debt is larger than the original loan. (2)

(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Monday 24 June 2024

Afternoon (Time: 1 hour 30 minutes)

Paper
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9FM0/4D

Further Mathematics

Advanced

PAPER 4D: Decision Mathematics 2

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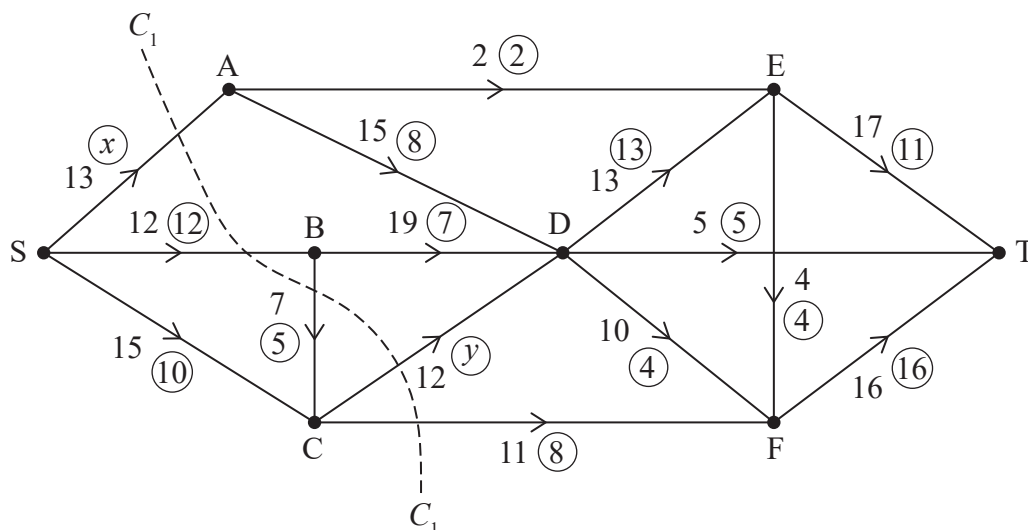


Figure 1

Figure 1 shows a capacitated, directed network of pipes. The numbers in circles represent an initial flow from S to T. The other number on each arc represents the capacity, in litres per second, of the corresponding pipe.

- (a) (i) State the value of x
(ii) State the value of y (2)

- (b) State the value of the initial flow. (1)

- (c) State the capacity of cut C_1 (1)

- (d) Find, by inspection, a flow-augmenting route to increase the flow by four units. You must state your route. (1)

The flow-augmenting route from (d) is used to increase the flow from S to T.

- (e) Prove that the flow is now maximal. (3)

A vertex restriction is now applied so that no more than 12 litres per second can flow through E.

- (f) (i) Complete Diagram 1 in the answer book to show this restriction.
(ii) State the **value** of the maximum flow through the network with this restriction. (2)

(Total for Question 1 is 10 marks)

2. The general solution of the first order recurrence relation

$$u_{n+1} + au_n = bn^2 + cn + d \quad n \geq 0$$

is given by

$$u_n = A(3)^n + 5n^2 + 1$$

where A is an arbitrary non-zero constant.

By considering expressions for u_{n+1} and u_n , find the values of the constants a , b , c and d .

(Total for Question 2 is 3 marks)

3. The table below shows the cost, in pounds, of transporting one unit of stock from each of four supply points, E, F, G and H, to three sales points, A, B and C. It also shows the stock held at each supply point and the amount required at each sales point. A minimum cost solution is required.

	A	B	C	Supply
E	23	28	22	21
F	26	19	29	32
G	29	24	20	29
H	24	26	19	23
Demand	45	19	23	

- (a) Explain why it is necessary to add a dummy demand point. (1)
- (b) On Table 1 in the answer book, insert appropriate values in the dummy demand column, D. (1)

After finding an initial feasible solution and applying one iteration of the stepping-stone method, the table becomes

	A	B	C	D
E	21			
F	19	13		
G		6	23	
H	5			18

- (c) Starting with GD as the next entering cell, perform two further iterations of the stepping-stone method to obtain an improved solution. You must make your method clear by showing your routes and stating the
- shadow costs
 - improvement indices
 - entering and exiting cells
- (6)
- (d) State the cost of the solution found in (c). (1)
- (e) Determine whether the solution obtained in (c) is optimal, giving a reason for your answer. (3)

(Total for Question 3 is 12 marks)

4. Four workers, A, B, C and D, are to be assigned to four tasks, P, Q, R and S.

Each task must be assigned to just one worker and each worker can do only one task.

Worker B cannot be assigned to task Q and worker D cannot be assigned to task R.

The amount, in pounds, that each worker would earn when assigned to each task is shown in the table below.

	P	Q	R	S
A	65	72	69	75
B	71	–	68	65
C	70	69	73	77
D	73	70	–	71

The Hungarian algorithm can be used to find the maximum total amount that would be earned by the four workers.

(a) (i) Explain how to modify the table so that the Hungarian algorithm could be applied.

(ii) Modify the table as described in (a)(i).

(3)

(b) Formulate the above situation as a linear programming problem. You must define the decision variables and make the objective function and constraints clear.

(6)

(Total for Question 4 is 9 marks)

5. Sebastien needs to make a journey. He can choose between travelling by plane, by train or by coach.

Sebastien knows the exact costs of all three travel options, but he also wants to account for his travel time, including any possible delays.

The cost of Sebastien's time is £50 per hour.

The table below shows the costs, the journey times (without delays), and the corresponding probabilities of delays, for each travel option.

	Cost of travel option	Journey time (in hours) without delays	Probability of a 1-hour delay	Probability of a 2-hour delay	Probability of a 3-hour delay	Probability of a 24-hour delay
Plane	£200	3	0.09	0.05	0	0.03
Train	£130	5	0.07	0.03	0	0
Coach	£70	6	0.15	0.1	0.05	0

- (a) By drawing a decision tree, evaluate the EMV of the total cost of Sebastien's journey for each node of your tree. (6)
- (b) Hence state the travel option that minimises the EMV of the total cost of Sebastien's journey. (1)
- (c) A cube root utility function is applied to the total costs of each option. Determine the travel option with the best expected utility and state the corresponding value. (3)

(Total for Question 5 is 10 marks)



6.

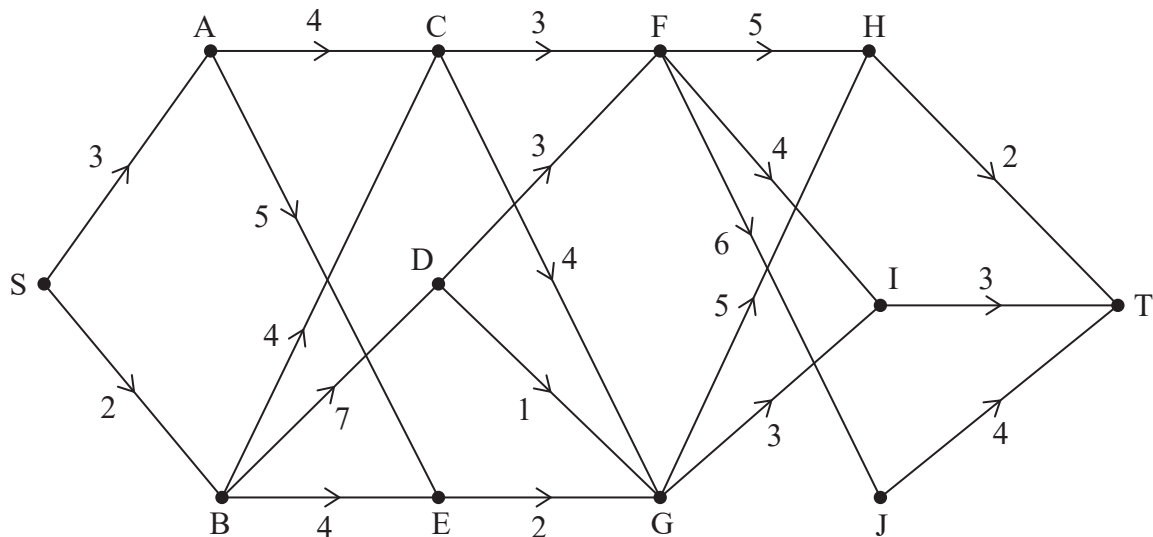


Figure 2

The staged, directed network in Figure 2 represents the roads that connect 12 towns, S, A, B, C, D, E, F, G, H, I, J and T. The number on each arc shows the time, in hours, it takes to drive between these towns.

Elena plans to drive from S to T. She must arrive at T by 9 pm.

- (a) By completing the table in the answer book, use dynamic programming to find the latest time that Elena can start her journey from S to arrive at T by 9 pm. (9)
- (b) Hence write down the route that Elena should take. (1)

(Total for Question 6 is 10 marks)

7.

		Player B		
		Option X	Option Y	Option Z
Player A	Option R	3	2	-3
	Option S	4	-2	1
	Option T	-1	3	6

A two person zero-sum game is represented by the pay-off matrix for player A, shown above.

(a) Verify that there is no stable solution to this game.

(2)

Player A intends to make a random choice between options R, S and T, choosing option R with probability p_1 , option S with probability p_2 and option T with probability p_3

Player A wants to find the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm.

Player A formulates the following objective function for the corresponding linear programme.

$$\text{Maximise } P = V \quad \text{where } V = \text{the value of the game} + 3$$

(b) Determine an initial Simplex tableau, making your variables and working clear.

(5)



After several iterations of the Simplex algorithm, a possible final tableau is

b.v.	V	p_1	p_2	p_3	r	s	t	u	Value
p_3	0	0	0	1	$\frac{1}{10}$	$-\frac{3}{80}$	$-\frac{1}{16}$	$\frac{33}{80}$	$\frac{33}{80}$
p_2	0	0	1	0	$-\frac{1}{10}$	$\frac{13}{80}$	$-\frac{1}{16}$	$\frac{17}{80}$	$\frac{17}{80}$
V	1	0	0	0	$\frac{1}{2}$	$\frac{5}{16}$	$\frac{3}{16}$	$\frac{73}{16}$	$\frac{73}{16}$
p_1	0	1	0	0	0	$-\frac{1}{8}$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$
P	0	0	0	0	$\frac{1}{2}$	$\frac{5}{16}$	$\frac{3}{16}$	$\frac{73}{16}$	$\frac{73}{16}$

(c) (i) State the best strategy for player A.

(ii) Calculate the value of the game for player B.

(3)

Player B intends to make a random choice between options X, Y and Z.

(d) Determine the best strategy for player B, making your method and working clear.

(3)

(Total for Question 7 is 13 marks)

8. A sequence $\{u_n\}$, where $n \geq 0$, satisfies the recurrence relation

$$2u_{n+2} + 5u_{n+1} = 3u_n + 8n + 2$$

- (a) Find the general solution of this recurrence relation.

(5)

A particular solution of this recurrence relation has $u_0 = 1$ and $u_1 = k$, where k is a positive constant. All terms of the sequence are positive.

- (b) Determine the value of k .

(3)

(Total for Question 8 is 8 marks)

TOTAL FOR PAPER IS 75 MARKS



Pearson Edexcel Level 3 GCE

Monday 26 June 2023

Afternoon (Time: 1 hour 30 minutes)

Paper
reference

9FM0/4D

Further Mathematics

Advanced

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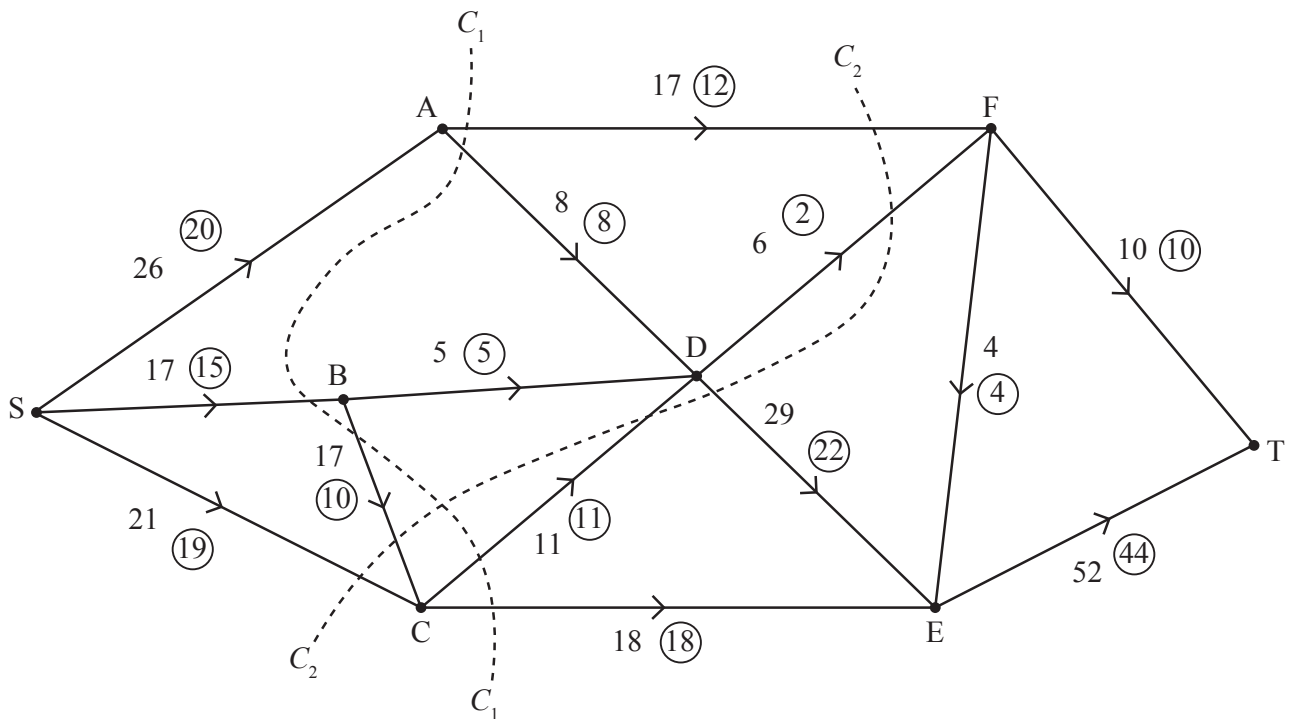


Figure 1

Figure 1 shows a capacitated, directed network of pipes. The number on each arc represents the capacity of that pipe. The numbers in circles represent a feasible flow from S to T.

- State the value of the feasible flow. (1)
- Find the capacity of cut C_1 and the capacity of cut C_2 (2)
- By inspection, find a flow-augmenting route to increase the flow by two units. You must state your route. (1)
- Prove that, once the flow-augmenting route found in (c) has been applied, the flow is now maximal. (3)

(Total for Question 1 is 7 marks)

2. An outdoor theatre is holding a summer gala performance. The theatre owner must decide whether to take out insurance against rain for this performance.

The theatre owner estimates that

- on a fine day, the total profit will be £15 000
- on a wet day, the total loss will be £20 000

Insurance against rain costs £2 000. If the performance must be cancelled due to rain, then the theatre owner will receive £16 000 from the insurer. If the performance is not cancelled due to rain, then the theatre owner will receive nothing from the insurer.

The probability of rain on the day of the gala performance is 0.2

Draw a decision tree and hence determine whether the theatre owner should take out the insurance against rain for this performance.

(5)

(Total for Question 2 is 5 marks)

3. The table below shows the stock held at each supply point and the stock required at each demand point in a standard transportation problem. The table also shows the cost, in pounds, of transporting the stock from each supply point to each demand point.

	Q	R	S	Supply
A	23	18	12	45
B	8	10	14	27
C	11	14	21	34
D	19	15	11	50
Demand	75	37	44	

The problem is partially described by the linear programming formulation below.

Let x_{ij} be the number of units transported from i to j

where $i \in \{A, B, C, D\}$

$j \in \{Q, R, S\}$ and $x_{ij} \geq 0$

$$\text{Minimise } P = 23x_{AQ} + 18x_{AR} + 12x_{AS} + 8x_{BQ} + 10x_{BR} + 14x_{BS} \\ + 11x_{CQ} + 14x_{CR} + 21x_{CS} + 19x_{DQ} + 15x_{DR} + 11x_{DS}$$

- Write down, as inequalities, the constraints of the linear program. (2)
- Use the north-west corner method to obtain an initial solution to this transportation problem. (1)
- Taking AS as the entering cell, use the stepping-stone method to find an improved solution. Make your route clear. (2)
- Perform one further iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by showing the route and the
 - shadow costs
 - improvement indices
 - entering cell and exiting cell
 (4)

(Total for Question 3 is 9 marks)



4. Four students, A, B, C and D, are to be allocated to four rounds, 1, 2, 3 and 4, in a competition. Each student is to take part in exactly one round and no two students may play in the same round.

Each student has been given an estimated score for each round. The estimated scores for each student are shown in the table below.

	1	2	3	4
A	34	20	18	15
B	49	31	12	34
C	48	27	23	26
D	52	45	42	42

- (a) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total estimated score. You must make your method clear and show the table after each stage.

(7)

- (b) Find this total estimated score.

(1)

(Total for Question 4 is 8 marks)

5. A sequence $\{u_n\}$, where $n \geq 0$, satisfies the second order recurrence relation

$$u_{n+2} = \frac{1}{2}(u_{n+1} + u_n) + 3 \quad \text{where } u_0 = 15 \quad u_1 = 20$$

(a) By considering the sequence $\{v_n\}$, where $u_n = v_n + 2n$ for $n \geq 0$, determine an expression for u_n as a function of n .

(7)

(b) Describe the long-term behaviour of u_n

(1)

(Total for Question 5 is 8 marks)



6. Polly is a motivational speaker who is planning her engagements for the next four weeks.

Polly will

- visit four different countries in these four weeks
- visit just one country each week
- leave from her home, S, and return there only after visiting the four countries
- travel directly from one country to the next

Polly wishes to determine a schedule of four countries to visit.

Table 1 shows the countries Polly could visit each week.

Week	1	2	3	4
Possible countries to visit	A or B	C, D or E	F or G	H, I or J

Table 1

Table 2 shows the speaker fee, in £100s, Polly would expect to earn in each country.

Country	A	B	C	D	E	F	G	H	I	J
Earnings in £100s	47	45	48	47	49	44	45	47	49	48

Table 2

Table 3 shows the cost, in £100s, of travelling between the countries.

	A	B	C	D	E	F	G	H	I	J
S	5	2						7	8	8
A			3	4	5					
B			5	4	6					
C						7	5			
D						6	7			
E						7	6			
F								6	7	8
G								7	8	6

Table 3

Polly's expected income is the value of the speaker fee minus the cost of travel.

She wants to find a schedule that maximises her total expected income for the four weeks.

Use dynamic programming to determine the optimal schedule. Complete the table provided in the answer book and state the maximum expected income.

(13)

(Total for Question 6 is 13 marks)

7. Martina decides to open a bank account to help her to save for a holiday. Each month she puts $\pounds k$ into the account and allows herself to spend one quarter of what was in the account at the end of the previous month.

Let u_n (where $n \geq 1$) represent the amount in the account at the end of month n .

Martina has $\pounds k$ in the account at the end of the first month.

- (a) By setting up a first order recurrence relation for u_{n+1} in terms of u_n , determine an expression for u_n in terms of n and k .

(6)

At the end of the 8th month, Martina needs to have at least $\pounds 1750$ in the account to pay for her holiday.

- (b) Determine, to the nearest penny, the minimum amount of money that Martina should put into the account each month.

(2)

(Total for Question 7 is 8 marks)



8. A two-person zero-sum game is represented by the pay-off matrix for player A shown below.

		Player B		
		Option X	Option Y	Option Z
Player A	Option Q	−3	2	5
	Option R	2	−1	0
	Option S	4	−2	−1
	Option T	−4	0	2

- (a) Verify that there is no stable solution to this game. (2)
- (b) Explain why player A should never play option T. You must make your reasoning clear. (2)

Player A intends to make a random choice between options Q, R and S, choosing option Q with probability p_1 , option R with probability p_2 and option S with probability p_3

Player A wants to calculate the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm.

- (c) (i) Formulate the game as a linear programming problem for player A. You should write the constraints as equations.
- (ii) Write down an initial Simplex tableau for this linear programming problem, making your variables clear. (7)

The linear programming problem is solved using the Simplex algorithm. The optimal value of p_1 is $\frac{6}{11}$ and the optimal value of p_2 is 0

- (d) Find the best strategy for player B, defining any variables you use. (6)

(Total for Question 8 is 17 marks)

TOTAL FOR PAPER IS 75 MARKS

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Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4D

Further Mathematics

Advanced

PAPER 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
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- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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P 7 2 1 1 8 A


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1. Four workers, A, B, C and D, are to be assigned to four tasks, 1, 2, 3 and 4. Each task must be assigned to just one worker and each worker must do only one task.

The cost of assigning each worker to each task is shown in the table below.

The total cost is to be minimised.

	1	2	3	4
A	32	45	34	48
B	37	39	50	46
C	46	44	40	42
D	43	45	48	52

- (a) Reducing rows first, use the Hungarian algorithm to obtain an allocation that minimises the total cost. You must make your method clear and show the table after each stage.

(5)

- (b) State the minimum total cost.

(1)

(Total for Question 1 is 6 marks)

2. The general solution of the second order recurrence relation

$$u_{n+2} + k_1 u_{n+1} + k_2 u_n = 0 \quad n \geq 0$$

is given by

$$u_n = (A + Bn)(-3)^n$$

where A and B are arbitrary non-zero constants.

- (a) Find the value of k_1 and the value of k_2

(2)

Given that $u_0 = u_1 = 1$

- (b) find the value of A and the value of B .

(2)

(Total for Question 2 is 4 marks)

3. The table below shows the transport options, usual travel times, possible delay times and corresponding probabilities of delay for a journey. All times are in minutes.

Transport option	Usual travel time	Possible delay time	Probability of delay
Car	52	10	0.10
		25	0.02
Train	45	15	0.05
		25	0.03
Coach	55	5	0.05
		15	0.01

- (a) Draw a decision tree to model the transport options and the possible outcomes. (5)
- (b) State the minimum expected travel time and the corresponding transport option indicated by the decision tree. (2)

(Total for Question 3 is 7 marks)

4.

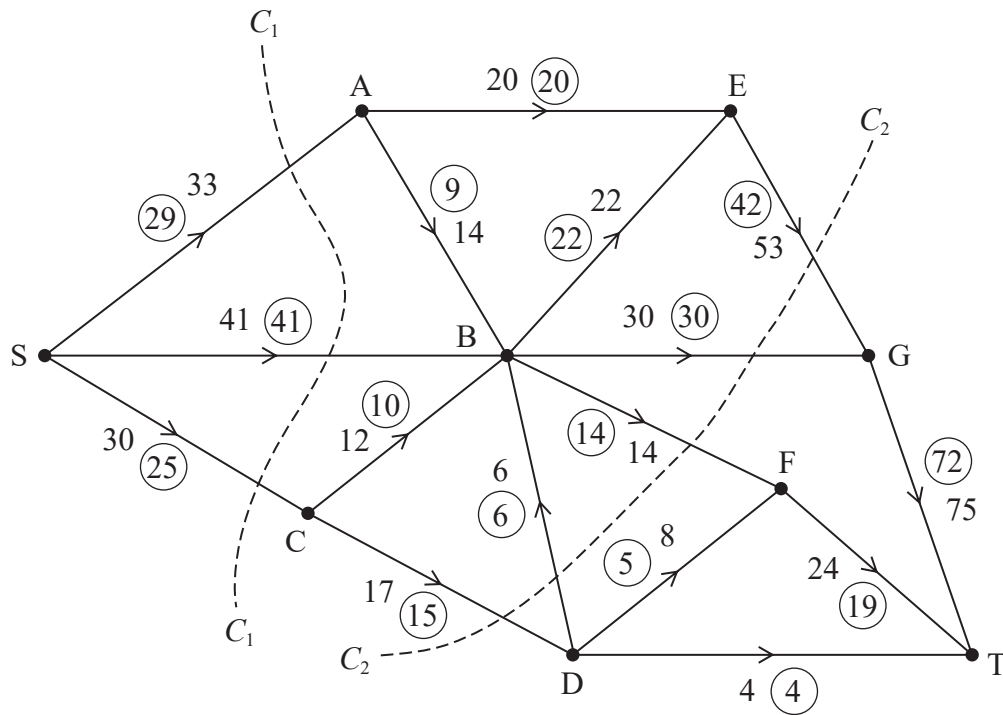


Figure 1

Figure 1 shows a capacitated, directed network of pipes. The uncircled number on each arc represents the capacity of the corresponding pipe. The numbers in circles represent an initial flow.

- List the saturated arcs. (1)
- State the value of the initial flow. (1)
- Explain why arc FT cannot be full to capacity. (1)
- State the capacity of cut C_1 and the capacity of cut C_2 . (2)
- By inspection find one flow-augmenting route to increase the flow by three units. You must state your route. (1)
- Prove that, once the flow-augmenting route found in part (e) has been applied, the flow is maximal. (3)

(Total for Question 4 is 9 marks)

5. A standard transportation problem is described in the linear programming formulation below.

Let x_{ij} be the number of units transported from i to j

where $i \in \{A, B, C, D\}$

$j \in \{R, S, T\}$ and $x_{ij} \geq 0$

Minimise $P = 23x_{AR} + 17x_{AS} + 24x_{AT} + 15x_{BR} + 29x_{BS} + 32x_{BT}$
 $+ 25x_{CR} + 25x_{CS} + 27x_{CT} + 19x_{DR} + 20x_{DS} + 25x_{DT}$

subject to

$$\sum x_{Aj} \leq 34$$

$$\sum x_{Bj} \leq 27$$

$$\sum x_{Cj} \leq 41$$

$$\sum x_{Dj} \leq 18$$

$$\sum x_{iR} \geq 44$$

$$\sum x_{iS} \geq 37$$

$$\sum x_{iT} \geq k$$

Given that the problem is balanced,

- (a) state the value of k . (1)
- (b) Explain precisely what the constraint $\sum x_{iR} \geq 44$ means in the transportation problem. (2)
- (c) Use the north-west corner method to obtain the cost of an initial solution to this transportation problem. (2)
- (d) Perform one iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by showing the route and the
- shadow costs
 - improvement indices
 - entering cell and exiting cell.
- (4)

(Total for Question 5 is 9 marks)

6. Bernie makes garden sheds. He can build up to four sheds each month.

If he builds more than two sheds in any one month, he must hire an additional worker at a cost of £250 for that month.

In any month in which sheds are made, the overhead costs are £35 for each shed made that month.

A maximum of three sheds can be held in storage at the end of any one month, at a cost of £80 per shed per month.

Sheds must be delivered at the end of the month.

The order schedule for sheds is

Month	January	February	March	April	May
Number ordered	1	3	3	5	2

There are no sheds in storage at the beginning of January and Bernie plans to have no sheds left in storage after the May delivery.

Use dynamic programming to determine the production schedule that minimises the costs given above. Complete the working in the table provided in the answer book and state the minimum cost.

(14)

(Total for Question 6 is 14 marks)

7.

		Player B			
		Option W	Option X	Option Y	Option Z
Player A	Option Q	4	3	-1	-2
	Option R	-3	5	-4	k
	Option S	-1	6	3	-3

A two person zero-sum game is represented by the pay-off matrix for player A shown above. It is given that k is an integer.

- (a) Show that Q is the play-safe option for player A regardless of the value of k . (2)

Given that Z is the play-safe option for player B,

- (b) determine the range of possible values of k . You must make your working clear. (2)

- (c) Explain why player B should never play option X. You must make your reasoning clear. (2)

Player A intends to make a random choice between options Q, R and S, choosing option Q with probability p_1 , option R with probability p_2 and option S with probability p_3

Player A wants to find the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm.

Given that $k > -4$, player A formulates the following objective function for the corresponding linear program.

Maximise $P = V$, where $V =$ the value of the original game $+ 4$

- (d) (i) Formulate the constraints of the linear programming problem for player A. You should write the constraints as equations.
(ii) Write down an initial Simplex tableau, making your variables clear. (7)

The Simplex algorithm is used to solve the linear programming problem. It is given that in the final Simplex tableau the optimal value of $p_1 = \frac{7}{37}$, the optimal value of $p_2 = \frac{17}{37}$ and all the slack variables are zero.

- (e) Determine the value of k , making your method clear. (4)

(Total for Question 7 is 17 marks)

8. The owner of a new company models the number of customers that the company will have at the end of each month. The owner assumes that
- a constant proportion, p (where $0 < p < 1$), of the previous month's customers will be retained for the next month
 - a constant number of new customers, k , will be added each month.

Let u_n (where $n \geq 1$) represent the number of customers that the company will have at the end of n months.

The company has 5000 customers at the end of the first month.

- (a) By setting up a first order recurrence relation for u_{n+1} in terms of u_n , determine an expression for u_n in terms of n , p and k .

(6)

The owner believes that 95% of the previous month's customers will be retained each month and that there will be 10 000 new customers each month.

According to the model, the company will first have at least 135 000 customers by the end of the m th month.

- (b) Using logarithms, determine the value of m .

(3)

(Total for Question 8 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS

Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/4D

Further Mathematics

Advanced

PAPER 4D: Decision Mathematics 2

You must have:

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Decision Mathematics Answer Book (enclosed)

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- There are 7 questions in this question paper. The total mark for this paper is 75.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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Pearson

1. Four workers, A, B, C and D, are to be assigned to three tasks, 1, 2 and 3. Each task must be assigned to just one worker and each worker can do one task only.

Worker A cannot do task 2 and worker D cannot do task 3

The cost of assigning each worker to each task is shown in the table below.

The total cost is to be minimised.

	1	2	3
A	53	–	62
B	48	57	59
C	55	63	58
D	69	49	–

Formulate the above situation as a linear programming problem. You must define your decision variables and make the objective function and constraints clear.

(6)

(Total for Question 1 is 6 marks)

2. Alka is considering paying £5 to play a game. The game involves rolling two fair six-sided dice. If the sum of the numbers on the two dice is at least 8, she receives £10, otherwise she loses and receives nothing.

If Alka loses, she can pay a further £5 to roll the dice again. If both dice show the same number then she receives £35, otherwise she loses and receives nothing.

- (i) Draw a decision tree to model Alka's possible decisions and the possible outcomes.
- (ii) Determine Alka's optimal EMV and state the optimal strategy indicated by the decision tree.

(7)

(Total for Question 2 is 7 marks)

3. The table below shows the cost, in pounds, of transporting one unit of stock from each of four supply points, A, B, C and D, to four sales points, P, Q, R and S. It also shows the number of units held at each supply point and the number of units required at each sales point.

A minimum cost solution is required.

	P	Q	R	S	Supply
A	18	19	17	13	28
B	16	15	14	19	43
C	21	17	22	23	29
D	16	20	19	21	36
Demand	25	41	40	30	

- (a) Use the north-west corner method to obtain an initial solution. (1)
- (b) Taking AS as the entering cell, use the stepping-stone method to find an improved solution. Make your method clear. (2)
- (c) Perform one further iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by showing the route and stating the
- shadow costs
 - improvement indices
 - entering cell and exiting cell
- (4)
- (d) State the cost of the solution found in (c). (1)
- (e) Determine whether the solution obtained in (c) is optimal, giving a reason for your answer. (3)

(Total for Question 3 is 11 marks)

4. Sequences $\{x_n\}$ and $\{y_n\}$ for $n \in \mathbb{N}$, are defined by

$$x_{n+1} = 2y_n + 3 \quad \text{and} \quad y_{n+1} = 3x_{n+1} - 4x_n$$

$$x_1 = 1 \quad \text{and} \quad y_1 = a$$

where a is a constant.

(a) Show that $x_{n+2} - 6x_{n+1} + 8x_n = 3$ (1)

(b) Solve the second-order recurrence relation given in (a) to obtain an expression for x_n in terms of a and n . (8)

Given that $x_7 = 28225$

(c) find the value of a . (2)

(Total for Question 4 is 11 marks)

5.

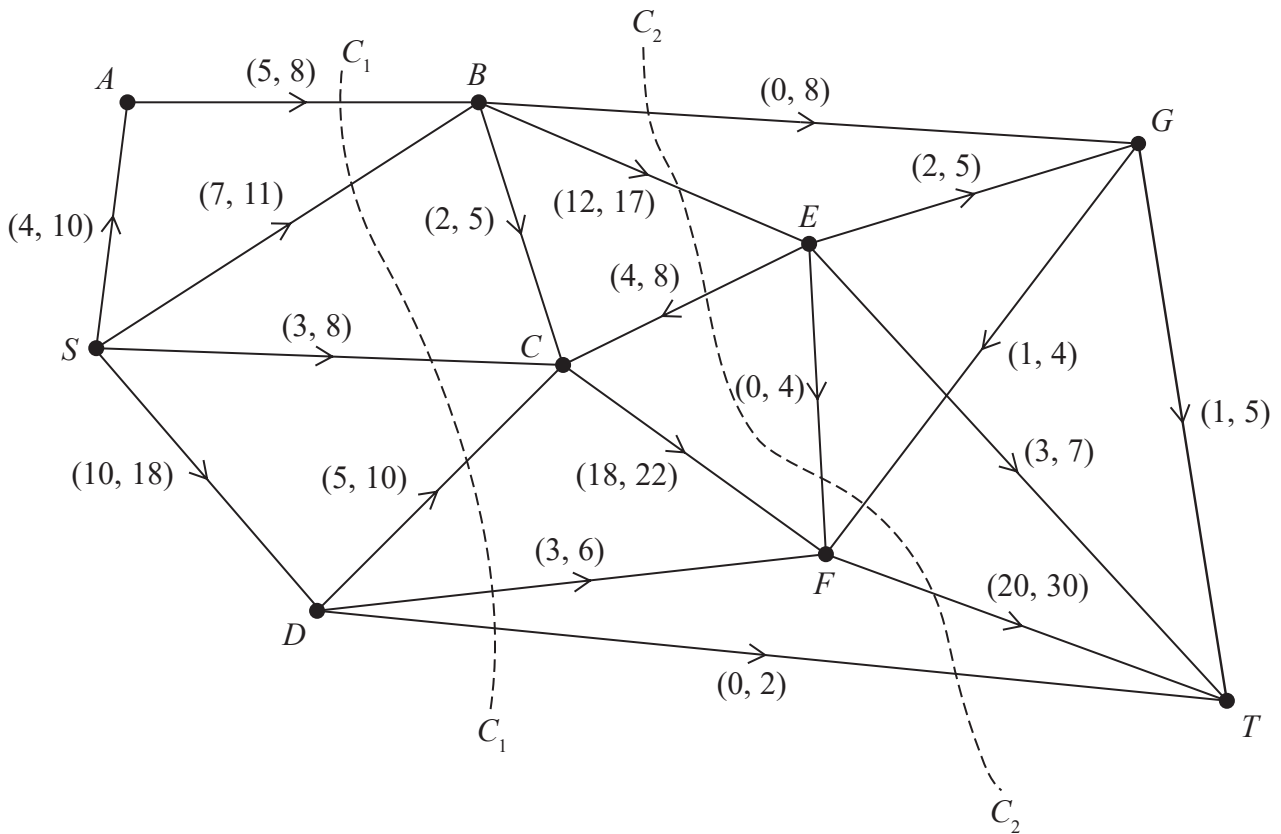


Figure 1

Figure 1 shows a capacitated, directed network. The network represents a system of pipes through which fluid can flow.

The weights on the arcs show the lower and upper capacities for the corresponding pipes, in litres per second.

(a) Calculate the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(b) Using only the capacities of cuts C_1 and C_2 , state what can be deduced about the maximum flow through the system.

(1)

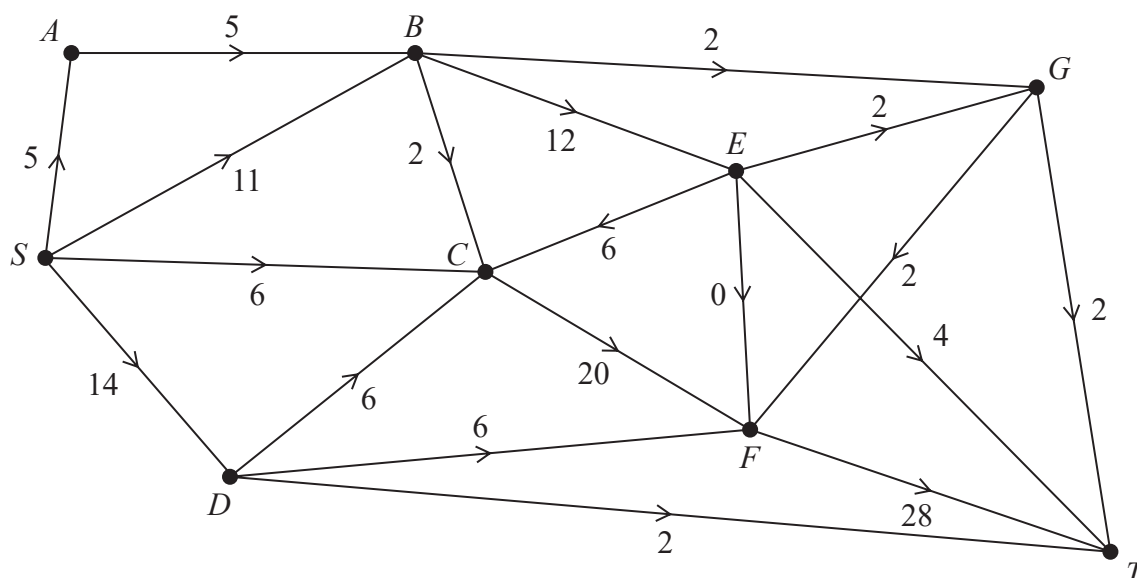


Figure 2

Figure 2 shows an initial flow through the same network.

- (c) State the value of the initial flow. (1)
- (d) By entering values along BC , CF and DT , complete the labelling procedure on Diagram 1 in the answer book. (2)
- (e) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow. (3)
- (f) Use your answer to (e) to find a maximum flow pattern for this system of pipes and draw it on Diagram 2 in the answer book. (1)
- (g) Prove that the answer to (f) is optimal. (3)
- A vertex restriction is now applied to B so that no more than 16 litres per second can flow through it.
- (h) (i) Complete Diagram 3 in the answer book to show this restriction. (3)
- (ii) State the value of the maximum flow with this restriction.

(Total for Question 5 is 16 marks)

6.

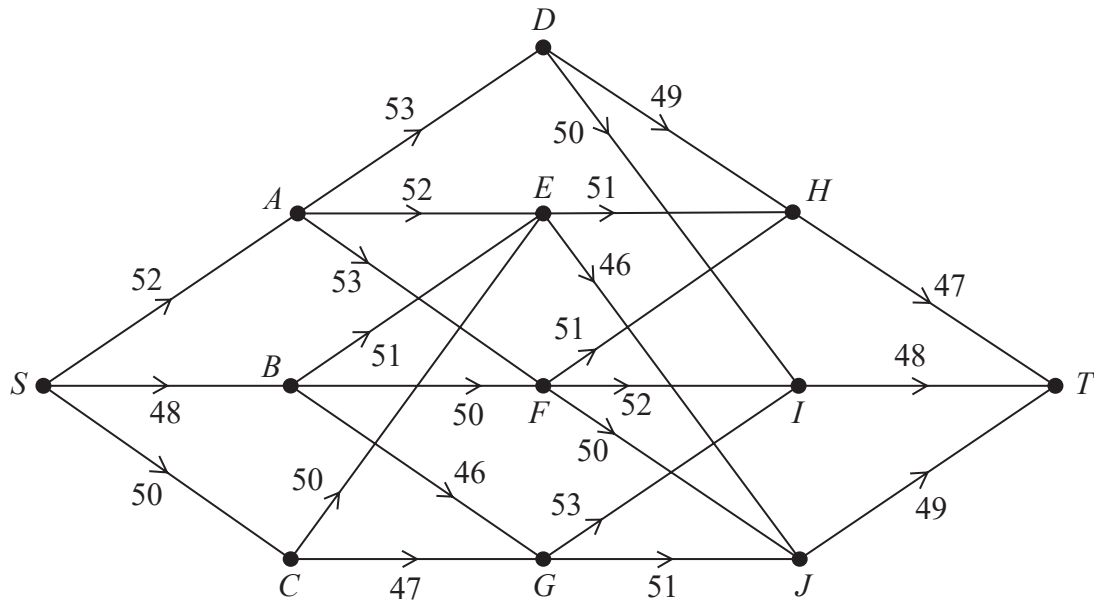


Figure 3

The staged, directed network in Figure 3 represents a series of roads connecting 12 towns, $S, A, B, C, D, E, F, G, H, I, J$ and T . The number on each arc shows the distance between these towns, in miles.

Bradley is planning a four-day cycle ride from S to T .

He plans to leave his home at S . On the first night he will stay at A, B or C , on the second night he will stay at D, E, F or G , on the third night he will stay at H, I or J , and he will arrive at his friend's house at T on the fourth day.

Bradley decides that the maximum distance he will cycle on any one day should be as small as possible.

- Write down the type of dynamic programming problem that Bradley needs to solve. (1)
- Use dynamic programming to complete the table in the answer book. (9)
- Hence write down the possible routes that Bradley could take. (2)

(Total for Question 6 is 12 marks)

7. Alexis and Becky are playing a zero-sum game.

Alexis has two options, Q and R. Becky has three options, X, Y and Z.

Alexis intends to make a random choice between options Q and R, choosing option Q with probability p_1 and option R with probability p_2

Alexis wants to find the optimal values of p_1 and p_2 and formulates the following linear programme, writing the constraints as inequalities.

$$\text{Maximise } P = V$$

where $V = 3 + \text{the value of the game to Alexis}$

$$\text{subject to } V \leq 6p_1 + p_2$$

$$V \leq 8p_2$$

$$V \leq 4p_1 + 2p_2$$

$$p_1 + p_2 \leq 1$$

$$p_1 \geq 0, p_2 \geq 0, V \geq 0$$

(a) Complete the pay-off matrix for Alexis in the answer book. (2)

(b) Use a graphical method to find the best strategy for Alexis. (6)

(c) Calculate the value of the game to Alexis. (1)

Becky intends to make a random choice between options X, Y and Z, choosing option X with probability q_1 , option Y with probability q_2 and option Z with probability q_3

(d) Determine the best strategy for Becky, making your method and working clear. (3)

(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS

Pearson Edexcel Level 3 GCE

Friday 23 October 2020

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

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- There are 7 questions in this question paper. The total mark for this paper is 75.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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P 6 2 6 8 2 R A


Pearson

1. Four workers, A, B, C and D, are to be assigned to four tasks, 1, 2, 3 and 4. Each worker must be assigned to exactly one task and each task must be done by exactly one worker.

Worker A cannot do task 3 and worker B cannot do task 4

The table below shows the profit, in pounds, that each worker would earn if assigned to each of the tasks.

	1	2	3	4
A	29	20	–	23
B	32	30	28	–
C	35	32	34	25
D	29	31	27	30

- (a) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total profit. You must make your method clear and show the table after each stage.

(7)

- (b) Determine the resulting total profit.

(1)

(Total for Question 1 is 8 marks)

2. Jenny can choose one of three options, A, B or C, when playing a game. The profit, in pounds, associated with each outcome and their corresponding probabilities are shown on the decision tree in Figure 1.

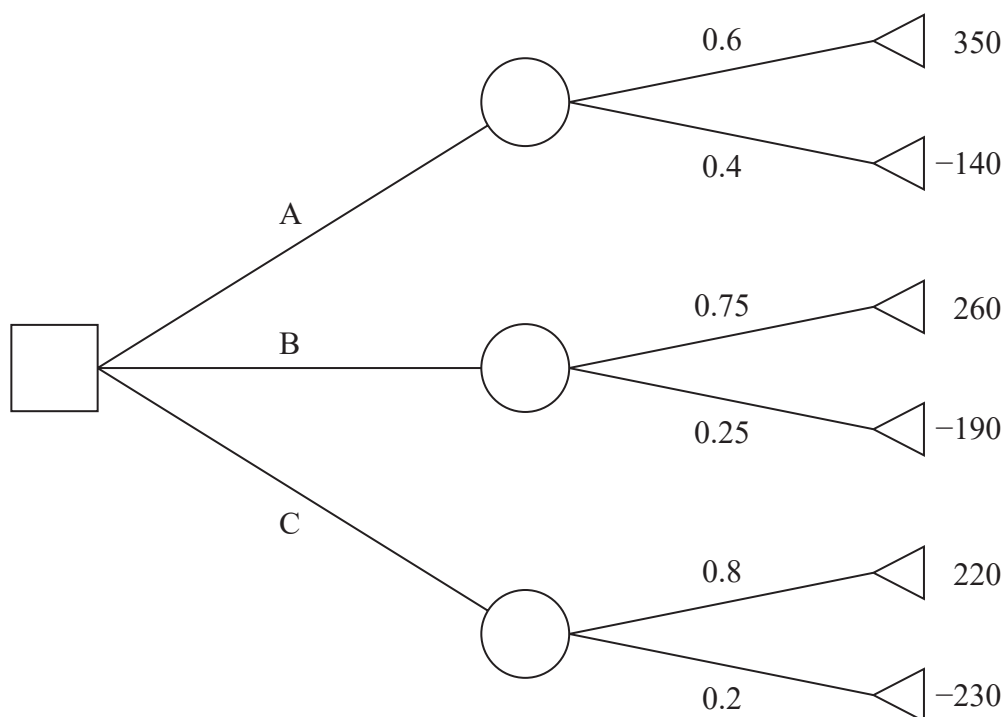


Figure 1

- (a) Calculate the optimal EMV to determine Jenny's best course of action. You must make your working clear. (3)

For a profit of £ x , Jenny's utility is given by $1 - e^{-\frac{x}{400}}$

- (b) Using expected utility as the criterion for the best course of action, determine what Jenny should do now to maximise her profit. You must make your working clear. (4)

(Total for Question 2 is 7 marks)

3. Table 1 shows the cost, in pounds, of transporting one unit of stock from each of four supply points, A, B, C and D, to three sales points, P, Q and R. It also shows the number of units held at each supply point and the number of units required at each sales point. A minimum cost solution is required.

	P	Q	R	Supply
A	25	24	17	42
B	7	12	14	68
C	13	11	20	25
D	16	15	13	40
Demand	59	72	44	

Table 1

Table 2 shows an initial solution given by the north-west corner method.

	P	Q	R
A	42		
B	17	51	
C		21	4
D			40

Table 2

- Taking AR as the entering cell, use the stepping-stone method to find an improved solution. Make your method clear. (2)
- Perform one further iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by stating
 - shadow costs
 - improvement indices
 - route
 - entering cell and exiting cell.(4)
- Determine whether the solution obtained from this second iteration is optimal, giving the reason for your answer. (3)
- Formulate this situation as a linear programming problem. You must define your decision variables and make the objective function and constraints clear. (6)
- Explain why the Simplex algorithm cannot be used to solve transportation linear programming problems such as that formulated in (d). (1)

(Total for Question 3 is 16 marks)

4. The complementary function for the second order recurrence relation

$$u_{n+2} + \alpha u_{n+1} + \beta u_n = 20(-3)^n \quad n \geq 0$$

is given by

$$u_n = A(2)^n + B(-1)^n$$

where A and B are arbitrary non-zero constants.

- (a) Find the value of α and the value of β .

(2)

Given that $2u_0 = u_1$ and $u_4 = 164$

- (b) find the solution of this second order recurrence relation to obtain an expression for u_n in terms of n .

(6)

(Total for Question 4 is 8 marks)

5.

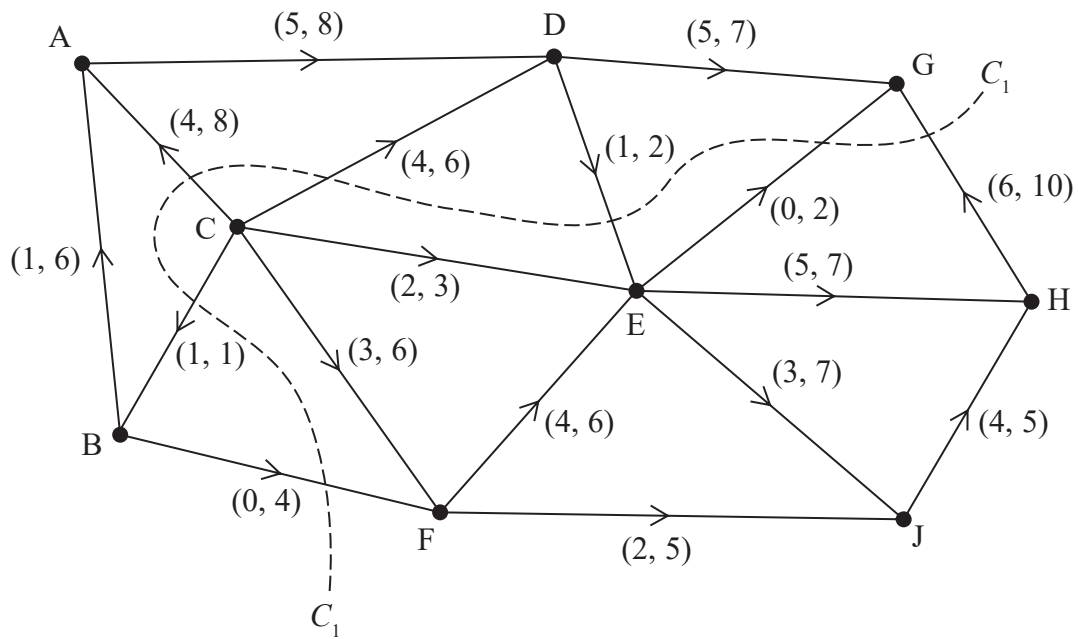


Figure 2

Figure 2 shows a capacitated, directed network. The network represents a system of pipes through which fluid can flow.

The weights on the arcs show the lower capacities and upper capacities for the corresponding pipes, in litres per second.

- State the source node. (1)
- Explain why the sink node must be G. (1)
- Calculate the capacity of the cut C_1 . (1)
- Assuming that a feasible flow exists,
 - explain why arc JH must be at its upper capacity,
 - explain why arcs AD and CD must be at their lower capacities. (2)
- Use Diagram 1 in the answer book to show a flow of 18 litres per second through the system. (2)
- Prove that the answer to (e) is the maximum flow through the system. (3)

(Total for Question 5 is 10 marks)

6.

		Player B		
		Option X	Option Y	Option Z
Player A	Option Q	1	5	3
	Option R	4	−3	1
	Option S	2	−4	−2
	Option T	3	−2	0

A two person zero-sum game is represented by the pay-off matrix for player A, shown above.

- (a) Explain, with justification, why this matrix may be reduced to a 3×3 matrix by removing option S from player A's choices. (2)
- (b) Verify that there is no stable solution to the reduced game. (3)

Player A intends to make a random choice between options Q, R and T, choosing option Q with probability p_1 , option R with probability p_2 and option T with probability p_3

Player A wants to find the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm. Player A formulates the following linear programme, writing the constraints as inequalities.

Maximise $P = V$, where $V = \text{the value of original game} + 3$

subject to $V \leq 4p_1 + 7p_2 + 6p_3$

$V \leq 8p_1 + p_3$

$V \leq 6p_1 + 4p_2 + 3p_3$

$p_1 + p_2 + p_3 \leq 1$

$p_1 \geq 0, p_2 \geq 0, p_3 \geq 0, V \geq 0$

- (c) Explain why V cannot exceed any of the following expressions

$$4p_1 + 7p_2 + 6p_3 \quad 8p_1 + p_3 \quad 6p_1 + 4p_2 + 3p_3 \quad (1)$$

- (d) Explain why it is necessary to use the constraint $p_1 + p_2 + p_3 \leq 1$ (1)

The Simplex algorithm is used to solve the linear programming problem.

Given that the optimal value of $p_1 = \frac{7}{11}$ and the optimal value of $p_3 = 0$

- (e) calculate the value of the game to player A. (3)

Player B intends to make a random choice between options X, Y and Z, choosing option X with probability q_1 , option Y with probability q_2 and option Z with probability q_3

- (f) Determine the optimal strategy for player B, making your working clear. (4)

(Total for Question 6 is 14 marks)

7. A manufacturer can export five batches of footwear each year. Each exported batch contains just one type of footwear. The types of footwear are trainers, sandals or high heels.

The table below shows the profit, in £1000s, for the number of batches of each type of footwear.

Number of batches	0	1	2	3	4	5
Trainers	0	50	90	170	225	295
Sandals	0	70	110	165	245	300
High heels	0	75	115	x	235	305

The total annual profit is to be maximised.

- (a) Use dynamic programming to determine the two possible values of the maximum total annual profit, giving one of these values in terms of x .
(10)

Given that the maximum total annual profit is £320 000

- (b) advise the manufacturer on the possible ways in which the five batches should be allocated.
(2)

(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Tuesday 23 June 2020

Afternoon (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

Answer Book

Do not return the question paper with the answer book

Total Marks

Turn over ►

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P 6 2 6 8 2 R A 0 1 2 0



Pearson

1.

	1	2	3	4
A	29	20	–	23
B	32	30	28	–
C	35	32	34	25
D	29	31	27	30

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				



Question 1 continued

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				

	1	2	3	4
A				
B				
C				
D				

Handwriting practice lines for the answer area.

(Total for Question 1 is 8 marks)



2.

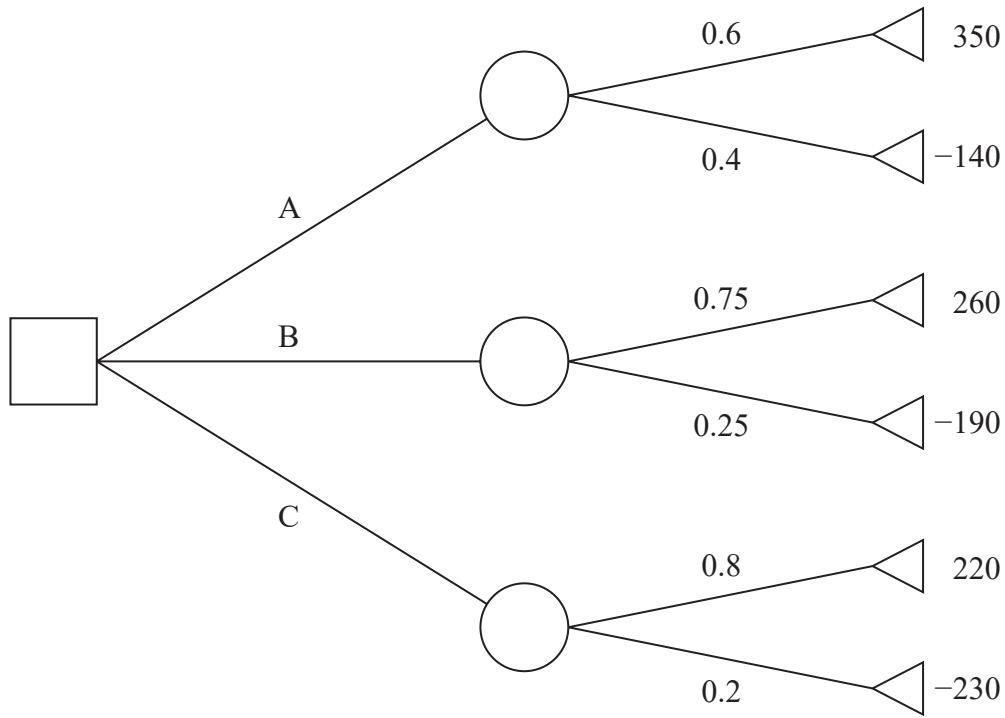


Figure 1

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Question 2 continued

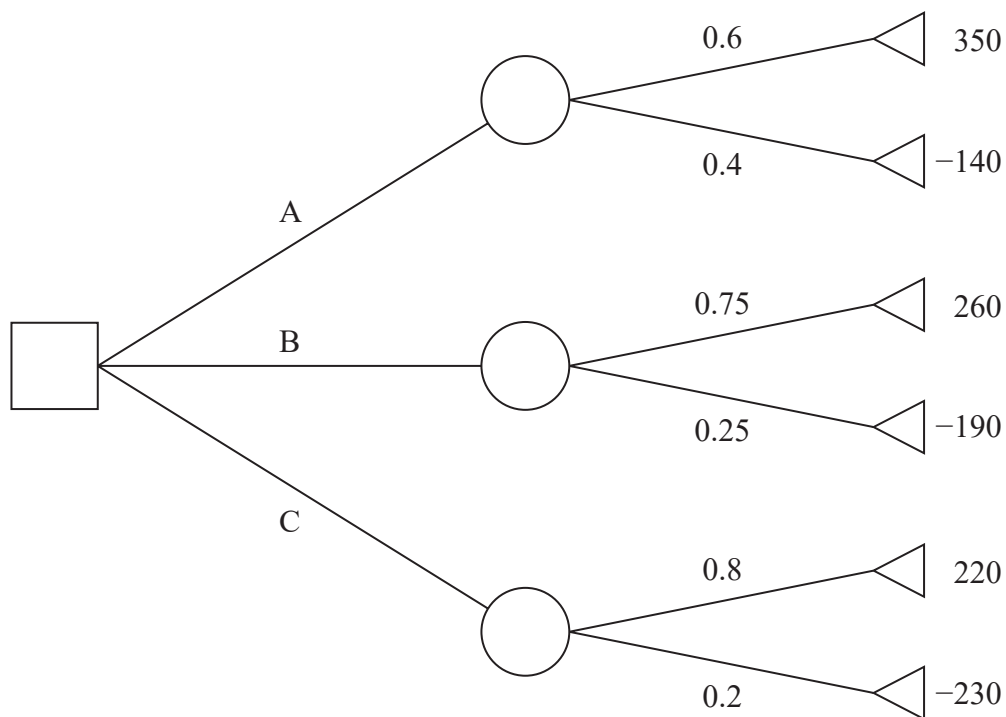


Figure 1

(Total for Question 2 is 7 marks)



P 6 2 6 8 2 R A 0 5 2 0

3.

	P	Q	R	Supply
A	25	24	17	42
B	7	12	14	68
C	13	11	20	25
D	16	15	13	40
Demand	59	72	44	

Table 1

	P	Q	R
A			
B			
C			
D			

	P	Q	R
A			
B			
C			
D			

	P	Q	R
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B			
C			
D			

	P	Q	R
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B			
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D			

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Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 16 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5.

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Question 5 continued

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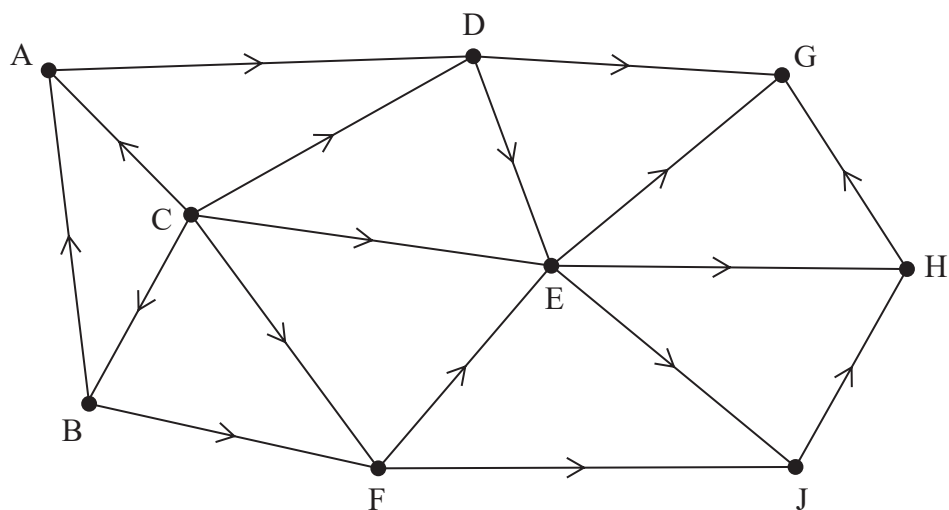


Diagram 1

(Total for Question 5 is 10 marks)



P 6 2 6 8 2 R A 0 1 3 2 0

6.

		Player B		
		Option X	Option Y	Option Z
Player A	Option Q	1	5	3
	Option R	4	-3	1
	Option S	2	-4	-2
	Option T	3	-2	0

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 14 marks)



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18

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Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Pearson Edexcel Level 3 GCE

Tuesday 25 June 2019

Morning (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read **each** question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the answer book provided.

1. Table 1 shows the cost, in pounds, of transporting one unit of stock from each of four supply points, A, B, C and D, to each of four demand points, P, Q, R and S. It also shows the stock held at each supply point and the stock required at each demand point. A minimum cost solution is required.

	P	Q	R	S	Supply
A	15	14	17	11	23
B	10	9	16	12	42
C	11	13	8	10	18
D	15	13	16	17	19
Demand	25	45	12	20	

Table 1

Table 2 shows an initial solution given by the north-west corner method.

	P	Q	R	S
A	23			
B	2	40		
C		5	12	1
D				19

Table 2

- (a) Taking DQ as the entering cell, use the stepping-stone method to find an improved solution. Make your method clear. (2)
- (b) Perform one further iteration of the stepping-stone method to obtain an improved solution. You must make your method clear by stating the
- shadow costs
 - improvement indices
 - route
 - entering cell and exiting cell.
- (4)
- (c) Determine whether the solution obtained from this second iteration is optimal, giving a reason for your answer. (3)
- (d) State the cost of the solution found in (b). (1)

(Total for Question 1 is 10 marks)

2. Four workers, Ted (T), Harold (H), James (J) and Margaret (M), are to be assigned to four tasks, 1, 2, 3 and 4. Each worker must be assigned to just one task and each task must be done by just one worker.

The profit, in pounds, resulting from allocating each worker to each task, is shown in the table below. The profit is to be maximised.

	1	2	3	4
T	103	97	74	80
H	201	155	145	155
J	111	80	77	92
M	203	188	137	184

- (a) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total profit. You must make your method clear and show the table after each stage.

(6)

- (b) Determine the resulting total profit.

(1)

(Total for Question 2 is 7 marks)

3.

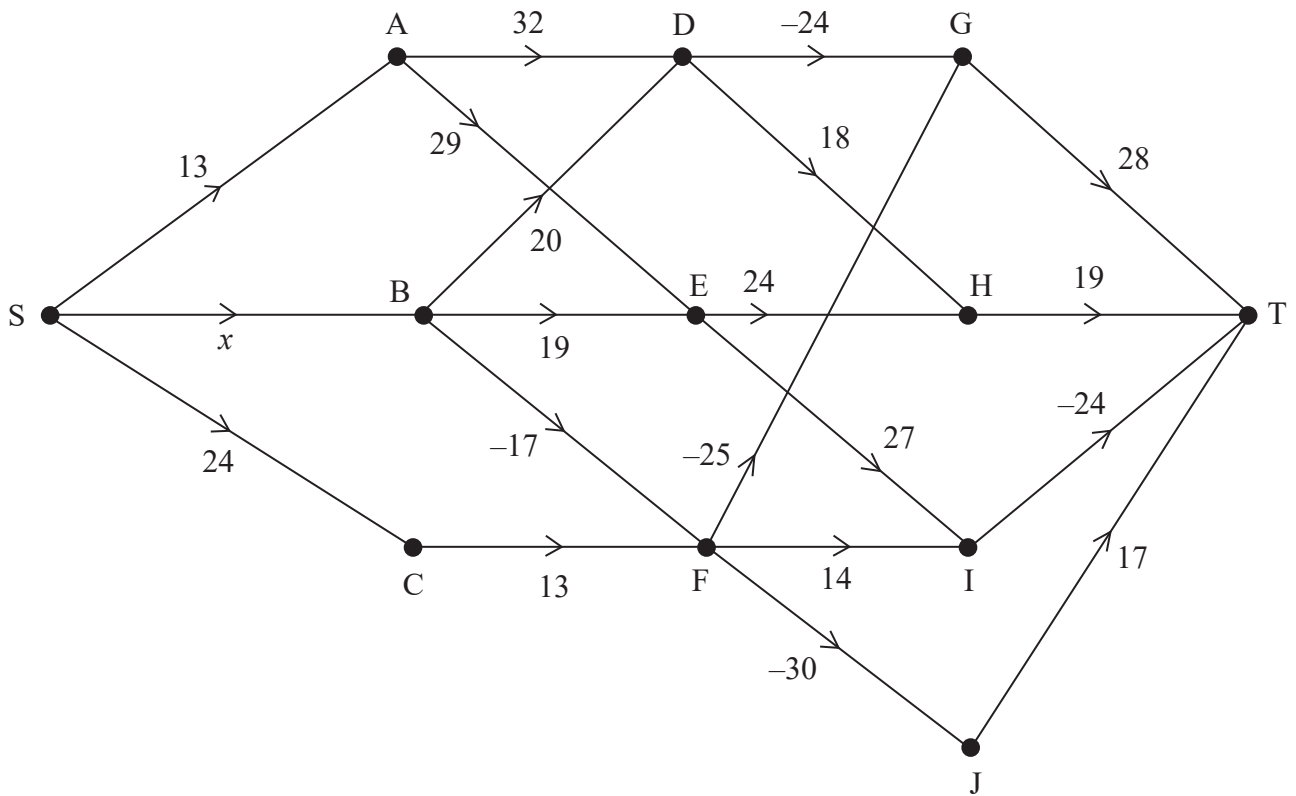


Figure 1

In Figure 1 the weight of arc SB is denoted by x where $x \geq 0$

- (a) Explain why Dijkstra's algorithm cannot be used on the directed network in Figure 1. (1)

It is given that the minimum weight route from S to T passes through B.

- (b) Use dynamic programming to find
- (i) the range of possible values of x
 - (ii) the minimum weight route from S to T. (12)

(Total for Question 3 is 13 marks)

4.

		Player B		
		Option X	Option Y	Option Z
Player A	Option P	3	-2	0
	Option Q	-4	4	-2
	Option R	1	2	-1

A two person zero-sum game is represented by the pay-off matrix for player A shown above.

(a) Verify that there is no stable solution to this game.

(2)

Player A intends to make a random choice between options P, Q and R, choosing option P with probability p_1 , option Q with probability p_2 and option R with probability p_3

Player A wants to find the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm. Player A formulates the following linear programming problem for the game, writing the constraints as inequalities.

Maximise $P = V$

subject to $V \geq 3p_1 - 4p_2 + p_3$

$V \geq -2p_1 + 4p_2 + 2p_3$

$V \geq -2p_2 - p_3$

$p_1 + p_2 + p_3 \leq 1$

$p_1 \geq 0, p_2 \geq 0, p_3 \geq 0, V \geq 0$

(b) Correct the errors made by player A in the linear programming formulation of the game, giving reasons for your answer.

(3)

(c) Write down an initial Simplex tableau for the corrected linear programming problem.

(3)

The Simplex algorithm is used to solve the corrected linear programming problem.

The optimal values are $p_1 = 0.6$, $p_2 = 0$ and $p_3 = 0.4$

(d) Calculate the value of the game to player A.

(2)

(e) Determine the optimal strategy for player B, making your reasoning clear.

(4)

(Total for Question 4 is 14 marks)

5. An increasing sequence $\{u_n\}$ for $n \in \mathbb{N}$ is such that the difference between the n th term of $\{u_n\}$ and the mean of the previous two terms of $\{u_n\}$ is always 6

(a) Show that, for $n \geq 3$

$$2u_n - u_{n-1} - u_{n-2} = 12 \quad (2)$$

Given that $u_1 = 2$ and $u_2 = 8$

(b) find the solution of this second order recurrence relation to obtain an expression for u_n in terms of n . (7)

(c) Show that as $n \rightarrow \infty$, $u_n \rightarrow kn$ where k is a constant to be determined. You must give reasons for your answer. (2)

(Total for Question 5 is 11 marks)

6.

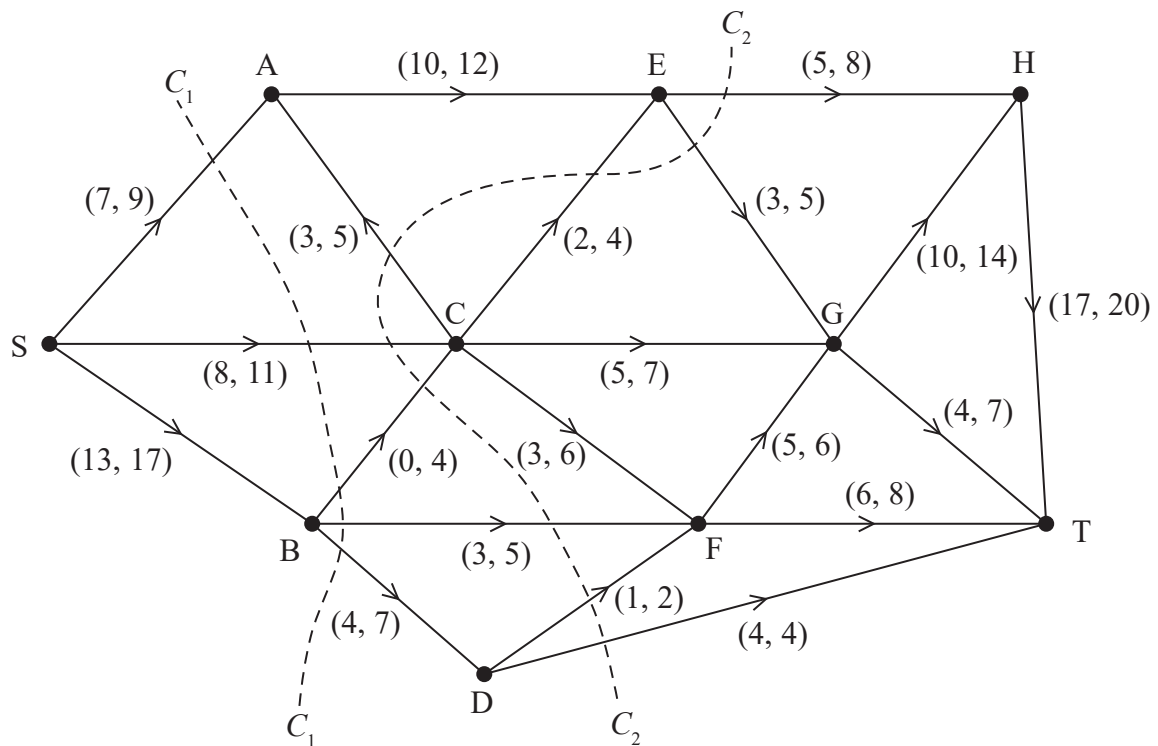


Figure 2

Figure 2 shows a capacitated, directed network. The network represents a system of pipes through which fluid flows from a source, S, to a sink, T.

The numbers (l, u) on each arc represent, in litres per second, the lower capacity, l , and the upper capacity, u , of the corresponding pipe.

Two cuts C_1 and C_2 are shown.

(a) Find the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(b) Explain why the arcs AE and CE cannot be at their upper capacities simultaneously.

(1)

(c) Explain why a flow of 31 litres per second through the system is not possible.

(1)

(d) Hence determine a minimum feasible flow and a maximum feasible flow through the system. You must draw these feasible flows on the diagrams in the answer book and give reasons to justify your answer. **You should not apply the labelling procedure to find these flows.**

(4)

(Total for Question 6 is 8 marks)

7. Aisha is deciding whether or not to play a game.

The game involves rolling three fair six-sided dice, which have faces numbered from 1 to 6

If the total score on the three dice is 16 or more then she wins a prize. If the total score is 15 or less then she loses and will have to pay the person running the game £3

(a) Given that the prize is £15

- (i) draw a decision tree to model Aisha's possible decisions and the possible outcomes
- (ii) determine Aisha's optimal EMV and state the optimal strategy indicated by the decision tree.

(6)

The utility function of the game to Aisha is $u(m) = 1 - e^{-\frac{m}{500}}$ where m is the amount of money that Aisha has available. Given that Aisha has exactly £3 and that the prize is now £ x

(b) find the expected utility to Aisha of playing the game in the form

$$\frac{a}{b} \left(1 - e^{-\frac{(x+c)}{500}} \right)$$

where a , b and c are integers to be found.

(2)

Aisha decides to use the expected utilities to determine whether she should play the game or not.

(c) Find the minimum prize for which Aisha would consider playing the game.

(4)

(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Tuesday 25 June 2019

Morning (Time: 1 hour 30 minutes)

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

Answer Book

Do not return the question paper with the answer book

Total Marks

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1.

	P	Q	R	S	Supply
A	15	14	17	11	23
B	10	9	16	12	42
C	11	13	8	10	18
D	15	13	16	17	19
Demand	25	45	12	20	

Table 1

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
A				
B				
C				
D				

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	P	Q	R	S
A				
B				
C				
D				

2.

	1	2	3	4
T	103	97	74	80
H	201	155	145	155
J	111	80	77	92
M	203	188	137	184

	1	2	3	4
T				
H				
J				
M				

	1	2	3	4
T				
H				
J				
M				

	1	2	3	4
T				
H				
J				
M				

	1	2	3	4
T				
H				
J				
M				

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	1	2	3	4
T				
H				
J				
M				

(a) _____

(a) _____

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4.

		Player B		
		Option X	Option Y	Option Z
Player A	Option P	3	-2	0
	Option Q	-4	4	-2
	Option R	1	2	-1

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Question 4 continued

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DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 14 marks)



5.

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

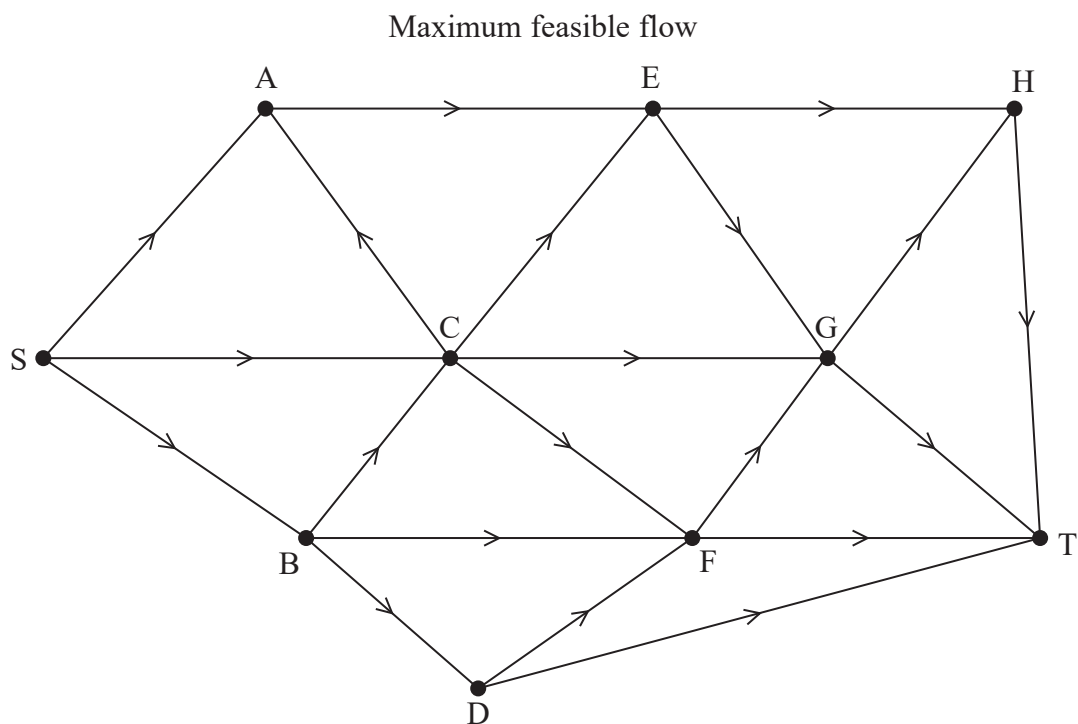
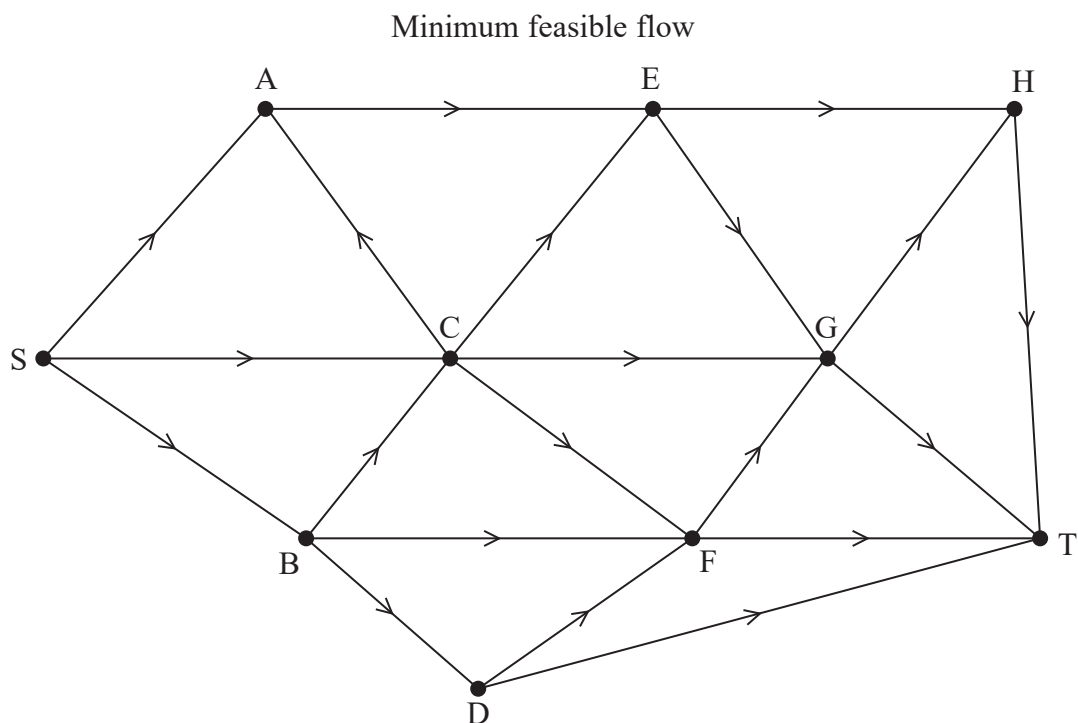
(Total for Question 5 is 11 marks)



6.

(a) (i) Capacity of cut $C_1 =$ _____(ii) Capacity of cut $C_2 =$ _____

Question 6 continued



(Total for Question 6 is 8 marks)



7.

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Pearson Edexcel Level 3 GCE

Mock Paper Set 1

Time: 1 hour 30 minutes

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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S 7 2 1 6 7 A



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1. Arpan is deciding which one of two investment schemes, X or Y, to choose.
- If Arpan chooses scheme X, he has a 90% chance of making a profit of £1400, but a 10% chance of making a loss of £300
 - If Arpan chooses scheme Y, he has an 80% chance of making a profit of £1250, but a 20% chance of making a loss of £150

For a profit of £ p , Arpan's utility is given by $\sqrt{p + 450}$

Draw a decision tree to help Arpan decide which scheme to choose to maximise his profit. Use expected utility as the criterion for the best course of action. You must make your method clear.

(6)

(Total for Question 1 is 6 marks)

2. The general solution of a first order linear non-homogeneous recurrence relation is given by

$$u_n = A(2)^n - 3n^2 + 1$$

where A is a constant.

Determine the recurrence relation for u_{n+1} in terms of u_n and n . You must simplify your answer.

(3)

(Total for Question 2 is 3 marks)

3. Four workers, A, B, C and D, are to be assigned to one of five tasks, P, Q, R, S and T.

Each worker must be assigned to at most one task, and each task must be done by at most one worker.

Worker C cannot be assigned to task S.

The amount, in pounds, that each worker will earn while assigned to each task is shown in the table below.

The Hungarian algorithm is to be used to find the maximum total amount that can be earned by the four workers.

	P	Q	R	S	T
A	48	53	47	50	51
B	44	41	45	39	40
C	45	49	41	–	44
D	51	52	53	51	55

- (a) (i) Explain how the table should be modified so that the Hungarian algorithm can be used to solve the problem.

(3)

- (ii) Modify the table so that the Hungarian algorithm may be used.

(1)

- (b) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total amount earned.

(6)

(Total for Question 3 is 10 marks)

4.

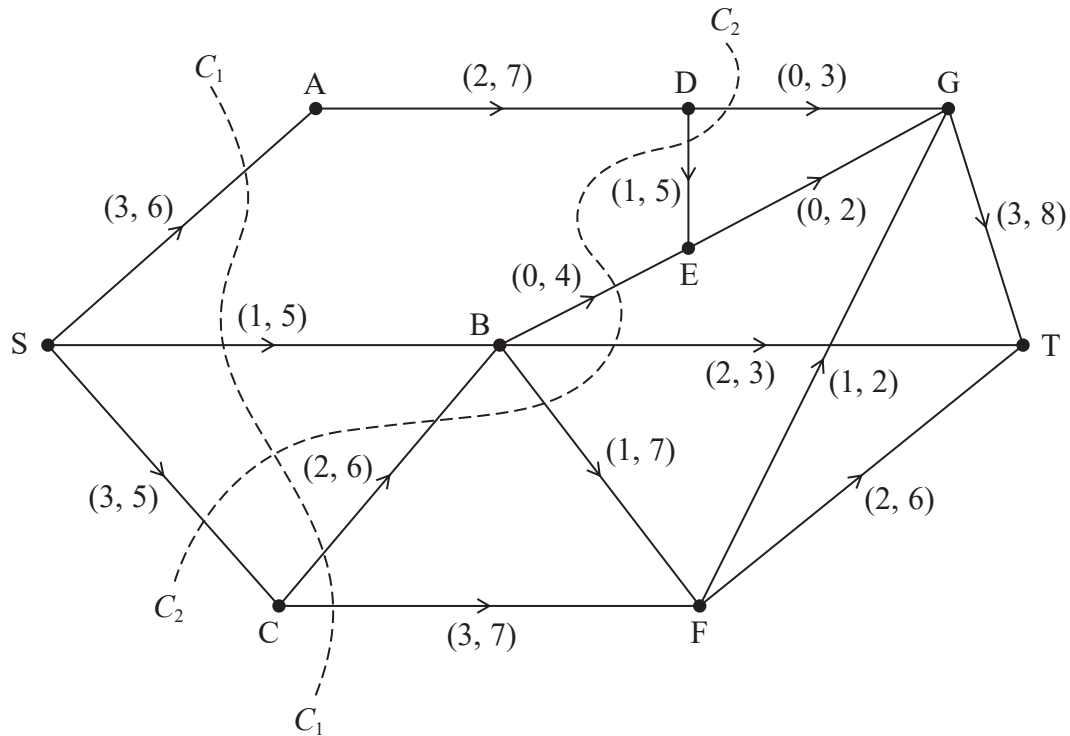


Figure 1

Figure 1 shows a capacitated, directed network. The network represents a system of pipes through which fluid can flow. The weights on the arcs show the lower and upper capacities for the corresponding pipes, in litres per second.

(a) Calculate the capacity of

(i) cut C_1

(ii) cut C_2

(2)

(b) Assuming that a feasible flow exists, explain why arcs CB and CF must both be at their lower capacities.

(1)

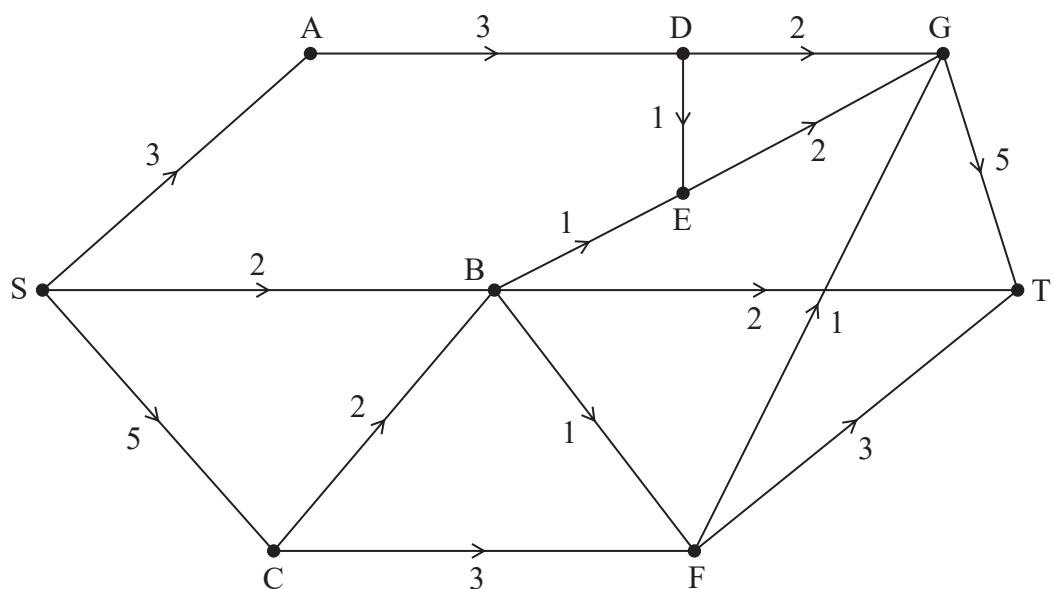


Figure 2

Figure 2 shows an initial flow through the same network.

- (c) By entering values along BE, CF and FT, complete the labelling procedure on Diagram 1 in the answer book. (2)
- (d) Use the labelling procedure to find a maximum flow through the network. You must state each flow-augmenting route you use, together with its flow. (3)
- (e) Use your answer to (d) to find a maximum flow pattern for this system of pipes and draw it on Diagram 2 in the answer book. (2)
- (f) Prove that the flow found in (e) is maximal. (3)

(Total for Question 4 is 13 marks)

5. The linear programming formulation below describes a standard transportation problem.

Let P be the total cost of transportation, in pounds.

Let x_{ij} be the number of units transported from supply point i to sales point j

where $i \in \{A, B, C\}$

$j \in \{Q, R, S, T\}$

$x_{ij} \geq 0$

Minimise $P = 175x_{AQ} + 150x_{AR} + 165x_{AS} + 180x_{AT}$

$+ 165x_{BQ} + 180x_{BR} + 190x_{BS} + 175x_{BT}$

$+ 175x_{CQ} + 155x_{CR} + 180x_{CS} + 165x_{CT}$

subject to $\sum x_{Aj} \leq 24$ $\sum x_{Bj} \leq 15$ $\sum x_{Cj} \leq 11$

$\sum x_{iQ} \geq 15$ $\sum x_{iR} \geq 14$ $\sum x_{iS} \geq 7$ $\sum x_{iT} \geq 14$

(a) Complete the table in the answer book to show

- the cost of transporting one unit of stock from each supply point to each sales point
- the number of units held at each supply point and the number of units required at each sales point

(2)

	Q	R	S	T
A	10	14		
B	5		7	3
C				11

The table above shows a solution to the linear programming problem.

(b) Taking AS as the entering cell, use the stepping-stone method once to find an improved solution. Make your working clear.

(2)

(c) Perform one further iteration of the stepping-stone method to obtain an improved solution. You must make your working clear by showing the route and

- the shadow costs
- the improvement indices
- the entering cell and the exiting cell

(4)

- (d) State the total cost of transportation for the solution found in (c). (1)
- (e) Determine whether the solution obtained in (c) is optimal, giving a reason for your answer. (3)

(Total for Question 5 is 12 marks)

6. A builder has a house to convert into four flats, A, B, C and D, at the rate of one flat every three months over the course of one year.

The conversion costs over the year will vary, depending on which flats have already been completed and which flat is currently being converted.

The conversion costs, in thousands of pounds, are shown in the table below.

Already converted	Conversion cost			
	A	B	C	D
None	40	50	35	40
A	–	35	40	35
B	45	–	50	40
C	35	45	–	55
D	45	45	50	–
A and B	–	–	60	30
A and C	–	45	–	50
A and D	–	50	60	–
B and C	60	–	–	50
B and D	55	–	45	–
C and D	70	60	–	–
A and B and C	–	–	–	60
A and B and D	–	–	45	–
A and C and D	–	55	–	–
B and C and D	60	–	–	–

The builder needs to choose the order in which the flats are converted.

The builder wants the greatest cost of each flat's conversion to be as small as possible.

- (a) Write down the type of dynamic programming problem that the builder needs to solve. (1)
- (b) Use dynamic programming to complete the table in the answer book. (9)
- (c) Hence, determine the order in which the four flats should be converted and the total cost of the conversions. (2)

(Total for Question 6 is 12 marks)

7. A particular solution for the second order recurrence relation

$$u_{n+2} + u_{n+1} - 6u_n = 3k^2(-3)^n \quad n \geq 0$$

where k is a non-zero constant is given by

$$u_n = kn(-3)^n$$

(a) Determine the general solution of the recurrence relation.

(5)

Given that $u_0 = u_1 = 0$

(b) find the value of u_{12}

(4)

(Total for Question 7 is 9 marks)

8.

		Player B		
		Option X	Option Y	Option Z
Player A	Option Q	3	-2	-5
	Option R	-2	-3	5
	Option S	1	-4	2
	Option T	-6	3	6

A two person zero-sum game is represented by the pay-off matrix for player A shown above.

(a) Verify that there is no stable solution to this game.

(2)

Player A intends to make a random choice between options Q, R, S and T, choosing option Q with probability p_1 , option R with probability p_2 , option S with probability p_3 and option T with probability p_4

Player A wants to find the optimal values of p_1, p_2, p_3 and p_4 using the Simplex algorithm.

Player A formulates the following objective function for the corresponding linear program.

$$\text{Maximise } P = V, \text{ where } V = v + 6$$

(b) (i) Define the variable v

(ii) Explain what has been achieved by adding 6 to v

(iii) Explain why this addition was necessary.

(3)

An optimal tableau for the linear programming problem, where r, s, t and u are slack variables, is given below.

b.v.	V	p_1	p_2	p_3	p_4	r	s	t	u	Value
p_4	0	0	$\frac{2}{7}$	0	1	$\frac{1}{14}$	$-\frac{1}{14}$	0	$\frac{5}{14}$	$\frac{5}{14}$
p_3	0	0	$\frac{65}{63}$	1	0	$-\frac{1}{21}$	$\frac{10}{63}$	$-\frac{1}{9}$	$\frac{2}{21}$	$\frac{2}{21}$
V	1	0	$\frac{23}{63}$	0	0	$\frac{19}{42}$	$\frac{41}{126}$	$\frac{2}{9}$	$\frac{235}{42}$	$\frac{235}{42}$
p_1	0	1	$-\frac{20}{63}$	0	0	$-\frac{1}{42}$	$-\frac{11}{126}$	$\frac{1}{9}$	$\frac{23}{42}$	$\frac{23}{42}$
P	0	0	$\frac{23}{63}$	0	0	$\frac{19}{42}$	$\frac{41}{126}$	$\frac{2}{9}$	$\frac{235}{42}$	$\frac{235}{42}$

(c) (i) State the values of p_1, p_2, p_3 and p_4

(ii) Find the value of the game to player B.

(2)

Player B intends to make a random choice between options X, Y and Z, choosing option X with probability q_1 , option Y with probability q_2 and option Z with probability q_3

(d) Determine the optimal strategy for player B, making your working clear.

(3)

(Total for Question 8 is 10 marks)

TOTAL FOR PAPER IS 75 MARKS

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Pearson Edexcel Level 3 GCE

Mock Paper Set 2

(Time: 1 hour 30 minutes)

Paper
reference

9FM0/4D

Further Mathematics

Advanced

PAPER 4D: Decision Mathematics 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator,
Decision Mathematics Answer Book (enclosed)

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A sequence $\{u_n\}$, where $n \geq 0$, satisfies the recurrence relation

$$u_{n+1} - 2u_n - 3 = 0$$

Given that $u_0 = 5$, solve the recurrence relation, giving u_n in terms of n

(Total for Question 1 is 3 marks)

2. Nicola is playing a card game.

She has four different cards and will play a different card in each of the four rounds.

Each card scores points for any one of four different categories.

The categories are Attack (A), Defence (D), Magic (M) and Special Ability (S).

For each round, Nicola must

- choose a card to play
- choose the category for which she will use the card

In the game, Nicola must

- play each of the four cards exactly once
- choose each category exactly once

Card 2 cannot be used for Defence and card 4 cannot be used for Magic.

Nicola wants to maximise her total points.

The number of points that each card will score if used for the different categories is shown in the table.

	A	D	M	S
Card 1	32	39	53	61
Card 2	34	—	47	64
Card 3	28	32	39	48
Card 4	25	36	—	59

- (a) Reducing rows first, use the Hungarian algorithm to obtain an allocation that maximises the total points scored. You must make your method clear and show the table after each stage.

(6)

- (b) Determine Nicola's maximum possible score.

(1)

(Total for Question 2 is 7 marks)

3. Table 1 shows the cost, in pounds, of transporting one unit of stock from each of three supply points, A, B and C, to four sales points, P, Q, R and S. It also shows the number of units held at each supply point and the number of units required at each sales point. A minimum cost solution is required.

	P	Q	R	S	Supply
A	28	12	19	16	28
B	31	28	23	19	33
C	18	21	22	28	18
Demand	16	20	26	17	

Table 1

Table 2 shows an initial solution given by the north-west corner method.

	P	Q	R	S
A	16	12		
B		8	25	
C			1	17

Table 2

- Taking CP as the entering cell, use the stepping-stone method once to find an improved solution. You must make your method clear. (2)
- Perform one further iteration of the stepping-stone method to obtain an improved solution.
You must make your method clear by showing the route and stating the
 - shadow costs
 - improvement indices
 - entering cell and exiting cell
 (4)
- State the cost of the solution found in part (b). (1)
- Determine whether the solution found in part (b) is optimal, giving the reason for your answer. (3)
- Formulate this situation as a linear programming problem. You must define your decision variables and make the objective function and constraints clear. (5)

(Total for Question 3 is 15 marks)

4. Weiyu is considering paying £1 to play a game.

The game involves selecting counters at random from a bag.

The bag contains 4 red, 3 green and 2 yellow counters.

Weiyu selects two counters at random from the bag, without replacement.

If she selects at least one yellow counter, then she wins £6, otherwise she receives nothing.

If Weiyu loses, she can then pay a further £1 to spin two fair spinners.

These spinners have four equal sectors numbered 1 to 4

The two spinners are spun and the sum of the numbers indicated on each spinner is recorded.

If the sum is more than 5, then she wins £8, otherwise she receives nothing.

- (a) Draw a decision tree to model Weiyu's possible decisions and the possible outcomes.

(6)

- (b) Determine Weiyu's optimal EMV, stating the optimal strategy indicated by the decision tree.

(2)

(Total for Question 4 is 8 marks)

5.

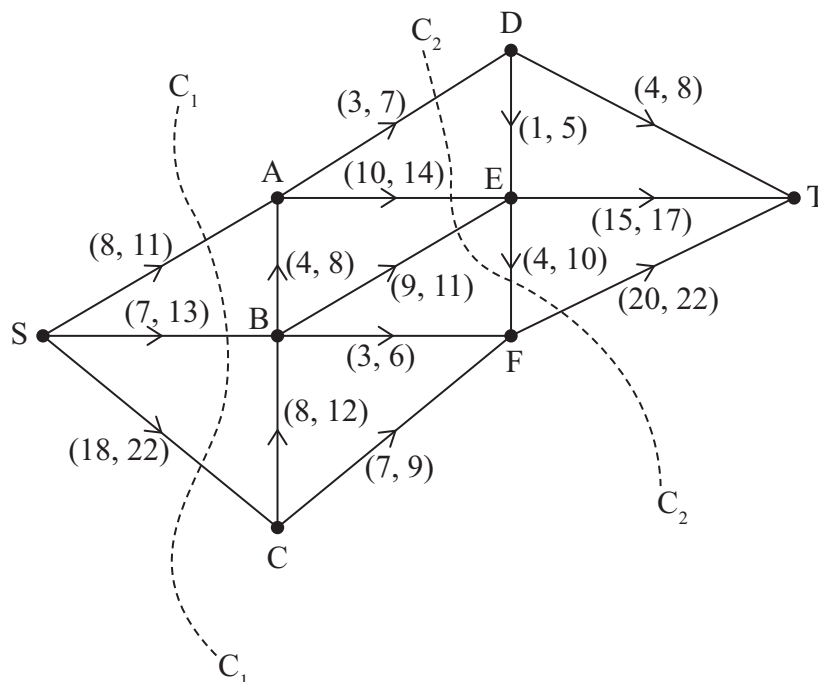


Figure 1

Figure 1 shows a capacitated, directed network. The network represents a system of pipes through which fluid can flow. The numbers on each arc represent lower and upper capacities for the corresponding pipes, in litres per second.

- (a) State the capacity of cut C_1 and the capacity of cut C_2 (2)
- (b) Using only the capacities of cuts C_1 and C_2 , state what can be deduced about the maximum flow through the system. (1)

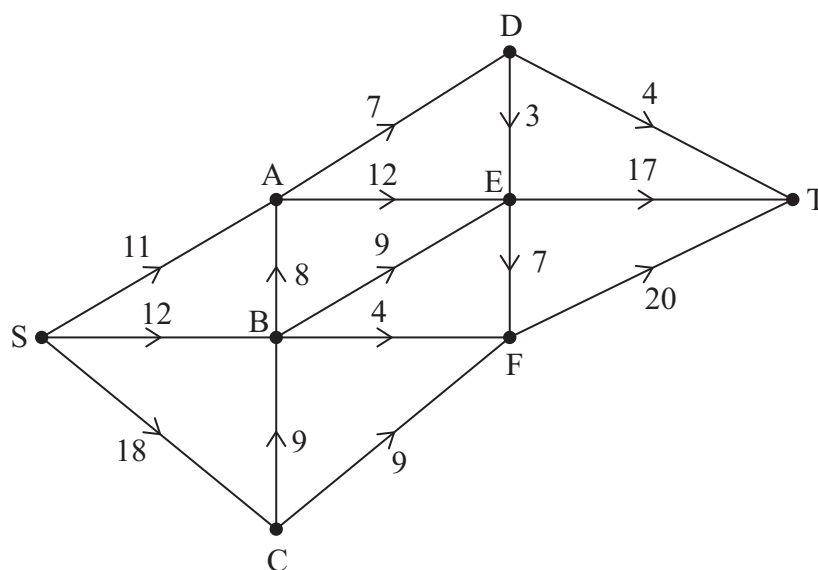


Figure 2

Figure 2 shows an initial flow through the same network.

- (c) Find the value of the initial flow. (1)
- (d) By entering values along CB, BF and FT, complete the labelling procedure on Diagram 1 in the answer book. (2)
- (e) Use the labelling procedure to find a maximum flow through the network. You must list each flow-augmenting route you use, together with its flow. (2)
- (f) Use the answer to part (e) to draw a maximum flow pattern for this system of pipes on Diagram 2 in the answer book. (1)
- (g) Prove that the answer to part (f) is optimal. (3)

A vertex restriction is now applied to B so that no more than 22 litres per second can flow through it.

- (h) (i) Complete Diagram 3 in the answer book to show this restriction. (3)
- (ii) State the value of the maximum possible flow with this restriction. (3)

(Total for Question 5 is 15 marks)

6.

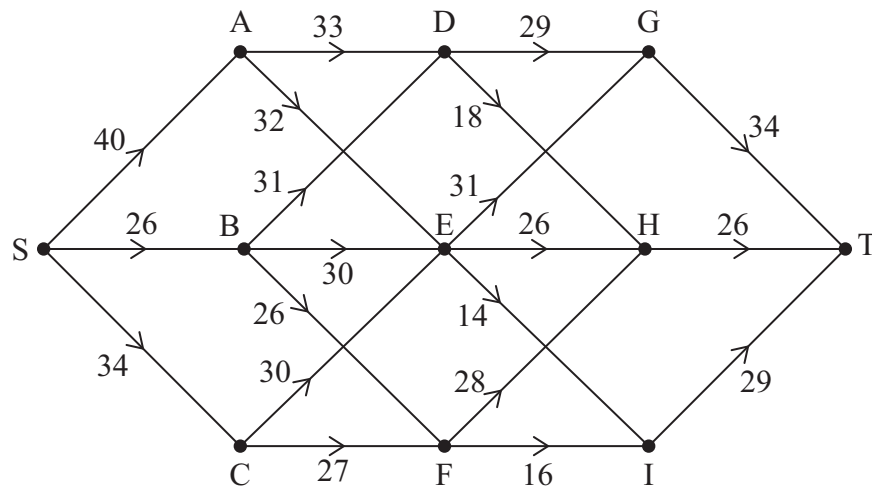


Figure 3

The directed network in Figure 3 represents the water slides in a theme park going from the top of the attraction (S) to the bottom (T).

The number on each arc represents the mean speed (in miles per hour) for each slide.

Navid wants to find a route from S to T so that the minimum speed he travels on any slide will be as large as possible.

- (a) Write down the type of dynamic programming problem that he needs to solve. (1)
- (b) Use dynamic programming to complete the table in the answer book and hence find the optimal route and the minimum speed Navid travels on any slide. (11)

(Total for Question 6 is 12 marks)

7.

		Player B		
		Option X	Option Y	Option Z
Player A	Option Q	2	6	x
	Option R	-1	-4	4
	Option S	-3	-5	1

A two-person zero-sum game is represented by the pay-off matrix for player A, where x is an integer.

- (a) Explain why player A should never play option S. You must make your reasoning clear.

(2)

Player A intends to make a random choice between options Q and R, choosing option Q with probability p

- (b) Given that the optimal value of p is $\frac{5}{8}$, determine the value of x , making your method clear.

(5)

(Total for Question 7 is 7 marks)

8. Shahzaib wants to build a row of wooden blocks in a straight line.
He has three different colours of blocks, red (R), green (G) and blue (B).

He can build 8 different rows of blocks of length 2 that do **not** have two adjacent blue blocks.

RR, RG, RB, GR, GG, GB, BR, BG

Let t_n be the number of distinct rows of blocks of length n that do **not** have two or more adjacent blue blocks.

- (a) Find the value of t_3 (1)
- (b) By setting up and solving a recurrence relation for t_{n+2} in terms of t_{n+1} and t_n , find the number of distinct rows of blocks of length 10 that do **not** have two or more adjacent blue blocks. (7)

(Total for Question 8 is 8 marks)

TOTAL FOR PAPER IS 75 MARKS

Pearson Edexcel Level 3 GCE

Further Mathematics

Advanced

Further Mathematics Option 2

Paper 4: Decision Mathematics 2

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/4G

You must have:

Decision Mathematics Answer Book (enclosed), calculator

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read **each** question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the answer book provided.

1. (a) Find the general solution of the recurrence relation

$$u_{n+2} = u_{n+1} + u_n, \quad n \geq 1 \quad (3)$$

Given that $u_1 = 1$ and $u_2 = 1$

- (b) find the particular solution of the recurrence relation.

(3)

(Total for Question 1 is 6 marks)

2.

	D	E	F	Available
A	15	19	9	25
B	11	18	10	55
C	11	12	18	20
Required	38	24	38	

A company has three factories, A, B and C. It supplies mattresses to three shops, D, E and F. The table shows the transportation cost, in pounds, of moving one mattress from each factory to each shop. It also shows the number of mattresses available at each factory and the number of mattresses required at each shop. A minimum cost solution is required.

(a) Use the north-west corner method to obtain an initial solution.

(1)

(b) Show how the transportation algorithm is used to solve this problem.

You must state, at each appropriate step, the

- shadow costs,
- improvement indices,
- route,
- entering cell and exiting cell,

and explain clearly how you know that your final solution is optimal.

(11)

(Total for Question 2 is 12 marks)

3. Four workers, A, B, C and D, are to be assigned to four tasks, P, Q, R and S.

Each worker must be assigned to at most one task and each task must be done by just one worker.

The amount, in pounds, that each worker would earn while assigned to each task is shown in the table below.

	P	Q	R	S
A	32	32	33	35
B	28	35	31	37
C	35	29	33	36
D	36	30	36	33

The Hungarian algorithm is to be used to find the maximum total amount which may be earned by the four workers.

- (a) Explain how the table should be modified. (1)
- (b) Reducing rows first, use the Hungarian algorithm to obtain an allocation which maximises the total earnings, stating how each table was formed. (7)
- (c) Formulate the problem as a linear programming problem. You must define your decision variables and make your objective function and constraints clear. (5)

(Total for Question 3 is 13 marks)

4. A game uses a standard pack of 52 playing cards.

A player gives 5 tokens to play and then picks a card. If they pick a 2, 3, 4, 5 or 6 then they gain 15 tokens. If any other card is picked they lose.

If they lose, the card is replaced and they can choose to pick again for another 5 tokens. This time if they pick either an ace or a king they gain 40 tokens. If any other card is picked they lose.

Daniel is deciding whether to play this game.

(a) Draw a decision tree to model Daniel's possible decisions and the possible outcomes. (6)

(b) Calculate Daniel's optimal EMV and state the optimal strategy indicated by the decision tree. (2)

(Total for Question 4 is 8 marks)

5.

	B plays 1	B plays 2	B plays 3	B plays 4
A plays 1	4	-2	3	2
A plays 2	3	-1	2	0
A plays 3	-1	2	0	3

A two person zero-sum game is represented by the pay-off matrix for player A given above.

(a) Explain, with justification, how this matrix may be reduced to a 3×3 matrix. (2)

(b) Find the play-safe strategy for each player and verify that there is no stable solution to this game. (4)

The game is formulated as a linear programming problem for player A.

The objective is to maximise $P = V$, where V is the value of the game to player A.

One of the constraints is that $p_1 + p_2 + p_3 \leq 1$, where p_1, p_2, p_3 are the probabilities that player A plays 1, 2, 3 respectively.

(c) Formulate the remaining constraints for this problem. Write these constraints as inequalities. (3)

The Simplex algorithm is used to solve the linear programming problem.

The solution obtained is $p_1 = 0, p_2 = \frac{3}{7}, p_3 = \frac{4}{7}$

(d) Calculate the value of the game to player A. (3)

(Total for Question 5 is 12 marks)

6.

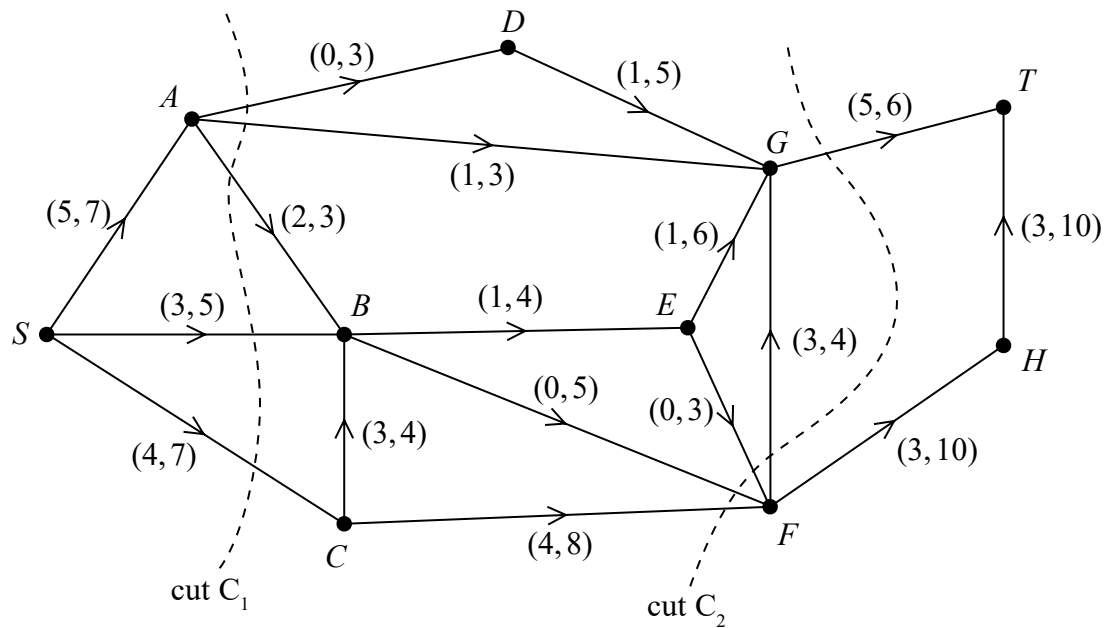


Figure 1

Figure 1 shows a capacitated, directed network. The number on each arc (x, y) represents the lower (x) capacity and upper (y) capacity of that arc.

- Calculate the value of the cut C_1 and cut C_2 (2)
 - Explain why the flow through the network must be at least 12 and at most 16 (1)
 - Explain why arcs DG, AG, EG and FG must all be at their lower capacities. (1)
 - Determine a maximum flow pattern for this network and draw it on Diagram 1 in the answer book. You do not need to use the labelling procedure. (2)
 - State the value of the maximum flow through the network.
 - Explain why the value of the maximum flow is equal to the value of the minimum flow through the network. (3)
- Node E becomes blocked and no flow can pass through it. To maintain the maximum flow through the network the upper capacity of exactly one arc is increased.
- Explain how it is possible to maintain the maximum flow found in (d). (3)

(Total for Question 6 is 12 marks)

7. A company assembles boats.

They can assemble up to five boats in any one month, but if they assemble more than three they will have to hire additional space at a cost of £800 per month.

The company can store up to two boats at a cost of £350 each per month.

The overhead costs are £1500 in any month in which work is done.

Boats are delivered at the end of each month. There are no boats in stock at the beginning of January and there must be none in stock at the end of May.

The order book for boats is

Month	January	February	March	April	May
Number ordered	3	2	6	3	4

Use dynamic programming to determine the production schedule which minimises the costs to the company. Show your working in the table provided in the answer book and state the minimum production cost.

(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS

Write your name here

Surname

Other names

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Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced

Further Mathematics Option 2

Paper 4: Decision Mathematics 2

Sample Assessment Material for first teaching September 2017

Paper Reference

9FM0/4G

Answer Book

Do not return the question paper with the answer book.

Total Marks

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Question 1 continued

(Total for Question 1 is 6 marks)

2.

	D	E	F	Available
A	15	19	9	25
B	11	18	10	55
C	11	12	18	20
Required	38	24	38	

(a)

	D	E	F	
A				
B				
C				

(b)

	D	E	F	
A				
B				
C				

	D	E	F	
A				
B				
C				

	D	E	F	
A				
B				
C				

	D	E	F	
A				
B				
C				

	D	E	F	
A				
B				
C				

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3. _____

	P	Q	R	S
A	32	32	33	35
B	28	35	31	37
C	35	29	33	36
D	36	30	36	33

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
A				
B				
C				
D				

	P	Q	R	S
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C				
D				

	P	Q	R	S
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B				
C				
D				

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5.

	B plays 1	B plays 2	B plays 3	B plays 4
A plays 1	4	-2	3	2
A plays 2	3	-1	2	0
A plays 3	-1	2	0	3

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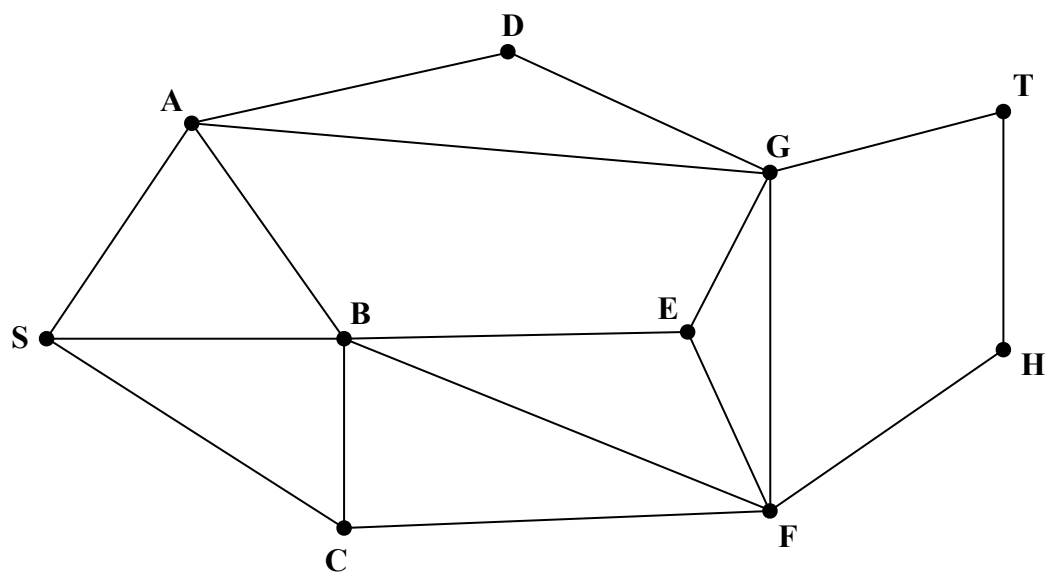
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Question 6 continued**Diagram 1****(Total for Question 6 is 12 marks)**

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Month	January	February	March	April	May
Number assembled					

Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS

Pearson Edexcel Level 3 GCE

Specimen Paper

Time: 1 hour 30 minutes

Paper Reference **9FM0/4D**

Further Mathematics

Advanced

Paper 4D: Decision Mathematics 2

You must have:

Decision Mathematics Answer Book (enclosed), calculator

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- Write your answers for this paper in the Decision Mathematics answer book provided.
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Do not return the question paper with the answer book.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read **each** question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the answer book provided.

1. Table 1 shows the cost, in pounds, of transporting one unit of stock from each of four supply points, A, B, C and D, to four demand points, 1, 2, 3 and 4. It also shows the stock held at each supply point and the stock required at each demand point. A minimum cost solution is required.

	1	2	3	4	Supply
A	23	32	25	30	15
B	34	29	32	28	17
C	37	27	31	33	23
D	34	28	29	35	16
Demand	18	20	14	19	

Table 1

Table 2 shows an initial solution given by the north-west corner method. Table 3 shows some of the improvement indices for this solution.

	1	2	3	4
A	15			
B	3	14		
C		6	14	3
D				16

Table 2

	1	2	3	4
A	X	14	3	6
B	X	X		
C		X	X	X
D	0	-1		X

Table 3

- (a) Calculate the shadow costs and enter the missing improvement indices into Table 3 in your answer book. (2)
- (b) Taking the most negative improvement index to indicate the entering cell, use the stepping-stone method once to obtain an improved solution. You must make your method clear by
- showing your route
 - stating the entering cell and exiting cell. (3)
- (c) Determine whether your current solution is optimal, giving a reason for your answer. (3)

The stock held at supply point A is increased to 20

For the transportation algorithm to be used, Table 1 needs to be modified.

- (d) (i) Explain why Table 1 needs to be modified.
- (ii) Describe how Table 1 should be modified.

You do not need to solve this problem.

(3)

(Total for Question 1 is 11 marks)

2. Four workers, A, B, C and D, are to be assigned to four tasks, 1, 2, 3 and 4. Each worker must be assigned to just one task and each task must be done by just one worker. The total cost of assigning the workers to the tasks is to be minimised.

The cost, in pounds, of assigning each worker to each task is shown in the table below.

	1	2	3	4
A	29	15	32	27
B	33	25	30	31
C	40	43	37	34
D	30	20	27	37

Part of a linear programming formulation to model this problem is as follows:

$$\text{Let } x_{ij} = \begin{cases} 1 & \text{if worker } i \text{ does task } j \\ 0 & \text{otherwise} \end{cases}$$

Where $i \in \{A, B, C, D\}$ and $j \in \{1, 2, 3, 4\}$

- (a) Write down the constraints of this linear programming problem. (2)
- (b) Write down the objective function of the linear programming problem. (1)
- (c) Reducing rows first, use the Hungarian algorithm to obtain an allocation that minimises the total cost of assigning all four workers to tasks. You must make your method clear and show the table after each stage. (6)

(Total for Question 2 is 9 marks)

3. Susie needs to buy a new boat. She has narrowed her choice down to one of three boats, A, B and C. From experience, Susie knows at the end of each year whichever boat she buys it will need repairing. For each of the three boats there is the option of buying annual insurance and, if bought, this insurance will cover all the repair costs at the end of the year. The cost, each year, of the insurance is £2000 for boat A, £2050 for boat B and £2100 for boat C.

If Susie does not buy insurance then she will have to pay the annual repair bill herself. At the end of the year the boat will require one of two types of repair. These two types can be classified as either 'minor' or 'major'. The repair costs and corresponding probabilities are shown in the table below.

Boat	Cost of minor repair	Probability of minor repair	Cost of major repair	Probability of major repair
A	1880	$\frac{4}{5}$	2350	$\frac{1}{5}$
B	2020	$\frac{3}{4}$	2300	$\frac{1}{4}$
C	1620	$\frac{2}{3}$	2670	$\frac{1}{3}$

- (a) Draw a decision tree to model Susie's possible decisions and the possible outcomes. (6)
- (b) Given that Susie wishes to minimise her potential expenditure, calculate her optimal EMV of annual repair costs and state the optimal strategy indicated by the decision tree. (2)

(Total for Question 3 is 8 marks)

4.

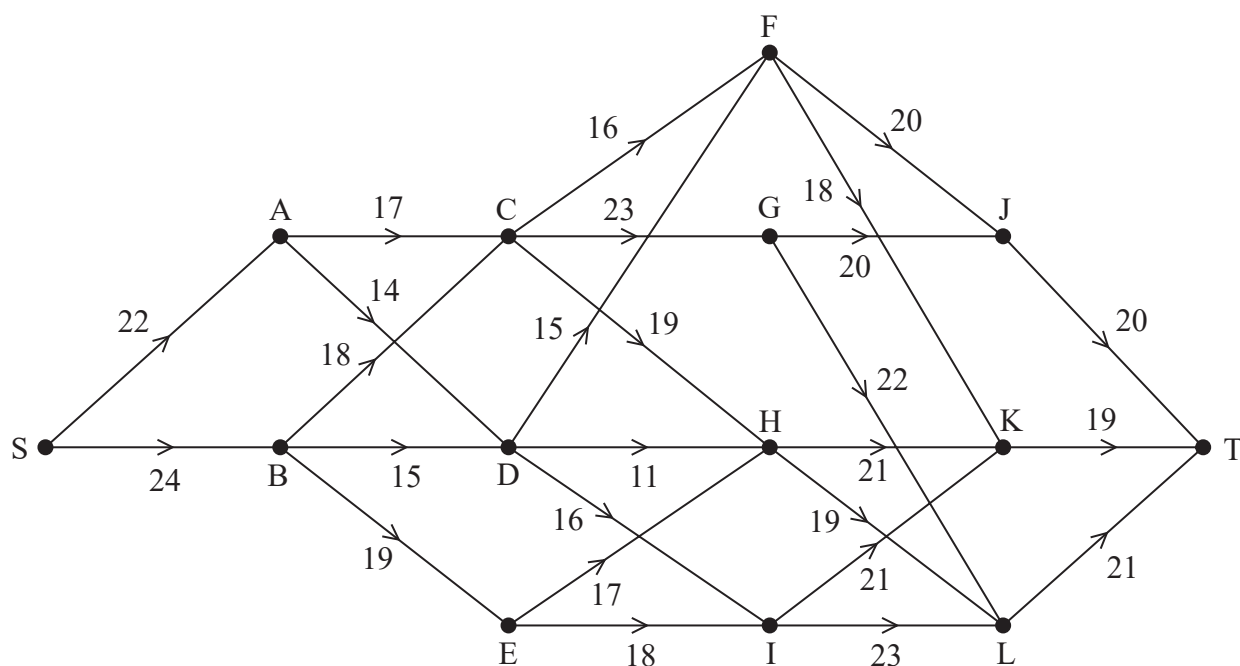


Figure 1

A company needs to transport goods from its base at S to its depot at T. The directed network shown in Figure 1 represents the roads from S to T. The number on each arc represents the maximum weight limit, in tonnes, for the corresponding road. The company needs to find the maximum possible weight of a loaded truck that can make the journey from S to T.

- (a) Write down the type of dynamic programming problem that the company needs to solve. (1)
- (b) Use dynamic programming to find
 - (i) the maximum weight of a loaded truck that can make the journey from S to T
 - (ii) the route that the truck could take. (12)

Due to engineering works, all roads leading out of B are closed.

- (c) (i) State the new maximum weight of a loaded truck that can make the journey from S to T.
- (ii) State the corresponding route. (2)

(Total for Question 4 is 15 marks)

5.

		Player B	
		Option X	Option Y
Player A	Option P	−3	3
	Option Q	2	1
	Option R	3	−2
	Option S	0	−1

A two person zero-sum game is represented by the pay-off matrix for player A shown above.

- Determine the number of points player B will gain if player A plays option R and player B plays option Y. (1)
- Explain, with justification, why this matrix may be reduced to a 3×2 matrix by removing option S from player A's choices. (2)
- Verify that there is no stable solution to this reduced game. (3)

Player A intends to make a random choice between her options P, Q and R, choosing option P with probability p_1 , option Q with probability p_2 and option R with probability p_3 .

She formulates the following linear programming problem so that she can find the optimal values of p_1 , p_2 and p_3 using the Simplex algorithm.

Maximise $P = V$

subject to $V \leq 5p_2 + 6p_3$

$V \leq 6p_1 + 4p_2 + p_3$

$p_1 + p_2 + p_3 \leq 1$

$p_1 \geq 0, p_2 \geq 0, p_3 \geq 0, V \geq 0$

- Show how she obtained the expressions

$$5p_2 + 6p_3 \quad \text{and} \quad 6p_1 + 4p_2 + p_3$$

- Explain why V cannot exceed either of these two expressions. (3)

- Explain why the constraint $p_1 + p_2 + p_3 \leq 1$ is needed. (1)

The Simplex algorithm is used to solve the linear programming problem.

The optimal value of p_1 is $\frac{1}{7}$ and the optimal value of p_3 is 0

- Calculate the optimal value of p_2
 - Calculate the value of the game to player B. (3)

(Total for Question 5 is 13 marks)

6. Sequences $\{x_n\}$ and $\{y_n\}$ for $n \in \mathbb{N}$, are defined by

$$x_{n+1} = 4y_n - 2 \quad \text{and} \quad y_{n+1} = 0.16x_n - 4$$

where $x_1 = -50.8$ and $y_1 = -11.52$

(a) Show that

$$x_{n+1} - 0.64x_{n-1} = k$$

where k is a constant to be determined.

(2)

(b) Find the solution of this second order recurrence relation to obtain an expression for x_n in terms of n .

(8)

(c) Hence find an expression for y_n in terms of n .

(2)

(d) State the limiting value of

(i) $\{x_n\}$

(ii) $\{y_n\}$

(1)

(Total for Question 6 is 13 marks)

7.

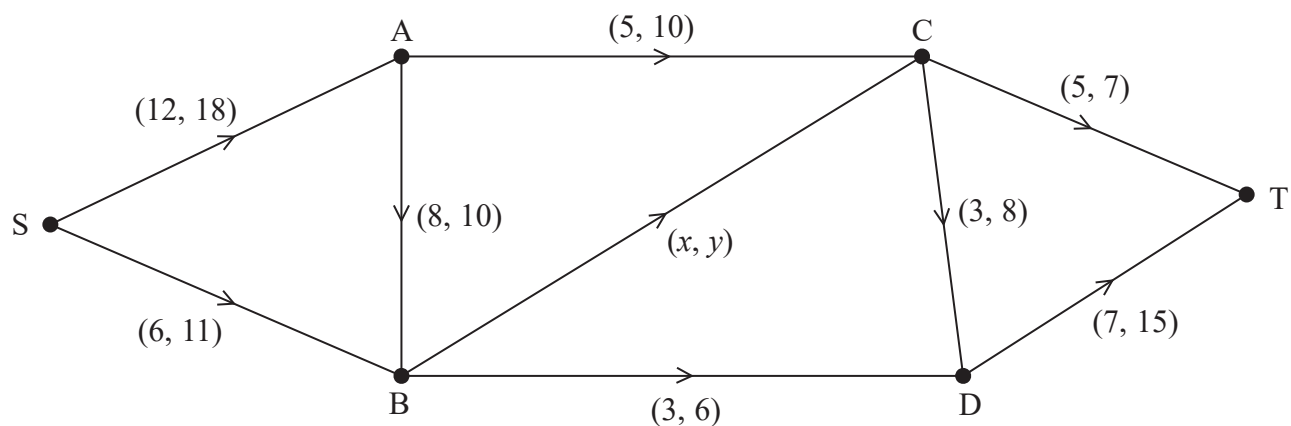


Figure 2

Figure 2 shows a capacitated, directed network. The numbers on each arc (l, u) represent the lower capacity (l) and upper capacity (u) of that arc.

- (a) Find the range of possible values of the flow in arc BC, giving reasons for your answer.

(4)

- (b) Write down the three constraints on the values of x and y .

(2)

(Total for Question 7 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS